

Research paper

APSYS: Automated Photometric Survey of Inactive Satellites for rotational dynamics and lightcurve characterization

Mădălina Trelia ^{a,b}, Dan Alin Nedelcu ^b, Mirel Birlan ^{b,c}

^a University of Bucharest, Faculty of Physics, Doctoral School of Physics, Atomistilor 405, Magurele, Ilfov, 077125, Romania

^b Astronomical Institute of the Romanian Academy, Cuștitul de Argint 5, Bucharest, 040557, Romania

^c IMCCE, Observatoire de Paris, 77 av Denfert Rochereau, Paris, 75014, France

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ABSTRACT

We present APSIS, the Automated Photometric Survey of Inactive Satellites, an automated, ground-based optical system for high-cadence photometric monitoring of inactive geosynchronous (GEO) and medium-Earth orbit (MEO) satellites. It integrates real-time image acquisition, automated calibration and reduction, and physically motivated light-curve modeling. Operating in two complementary modes, nightly GEO-belt surveys and targeted follow-up tracking, APSIS continuously updates a database of rotational states for a large set of objects. Since 2023, APSIS has obtained multi-epoch photometry for nine inactive GEO satellites, revealing near-annual periodic variations in rotation period. Fitting a torque-driven model yields families of admissible spin-axis orientations, demonstrating quantitative consistency between Solar Radiation Pressure forcing and the observed dynamics, although intrinsic symmetries prevent unique pole determinations. In parallel, we developed a minimal-parameter, physically motivated lightcurve inversion framework, modeling satellites as a diffuse-reflecting rectangular bus (Lommel–Seeliger law) and specular elements. A two-stage optimization (Differential Evolution and Levenberg–Marquardt refinement) applied to multiple epoch lightcurves of Raduga 3 (1977-071A) successfully reproduces key asymmetries, recovering a glinting panel nearly perpendicular to the spin axis and bus axis ratios consistent with known KAUR-3 dimensions. Extending the model to four glinting panels further reduced residuals and revealed a symmetric arrangement of reflective surfaces, consistent with paired solar arrays or radiators. These results highlight the capability of APSIS to deliver high-fidelity, long-term photometric monitoring and physically interpretable models of satellite geometry, providing a new pathway for characterizing inactive spacecraft in GEO and MEO regimes.

1. Introduction

The exponential development in space activity has led to a dramatic rise in the number of objects orbiting Earth, outpacing the regulatory and technological frameworks designed to ensure the long-term sustainability of near-Earth space. It is important to note that huge projects of the low-Earth orbit (LEO) satellite constellation dramatically changed the paradigm of space activities and space exploitation. As of 2022, approximately 80 % of all cataloged resident space objects (RSO) reside in LEO [1]. Although the deployment of mega-constellations as Starlink or OneWeb is of benefit to our society, it can also impact space activities, including astronomical observations and in-orbit spacecraft safety [1], significantly increasing the number of potential collisions of the later abandoned spacecraft with the operational satellites by 10 % [2].

However, the growing threat to orbital sustainability is not limited to operational satellites. Non-functional spacecrafts, space debris

resulting from accidental collisions [3,4], fragmentation [4] and unfortunately intentional destruction via anti-satellite (ASAT) tests [5,6], significantly outnumber active payloads. As of December 2022, debris generated solely by breakup events accounts for nearly 47 % of the total cataloged RSO [4]. Alongside, we may notice the self-destruction of satellites after their active period due to the uncontrolled rotation or a collision with micrometeorites and space debris [7].

1.1. GEO region — a contested and congested resource

Despite being less densely populated than LEO, the GEO belt presents particular challenges for long-term sustainability. Objects placed in GEO are not naturally removed from orbit, making any loss of control potentially permanent. An European Space Agency (ESA) report classifying objects in geosynchronous orbits shows that, as of 2017, over half (53 %) of the 1523 known objects were in drift orbits,

* Corresponding author at: Astronomical Institute of the Romanian Academy, Cuștitul de Argint 5, Bucharest, 040557, Romania.

E-mail addresses: madalina.trelia@astro.ro (M. Trelia), nedelcu@astro.ro (D.A. Nedelcu), birlan@astro.ro (M. Birlan).

with only 34 % being actively controlled [8]. Photometric observations further revealed that more than 50 % of observed GEO objects exhibit rotational motion, indicating significant non-cooperative behavior in the region [9].

ESA's 2025 Space Environment Report further highlights the persistent dominance of >1 000 kg payloads in GEO launches, although a trend toward launching smaller spacecraft (≤ 100 kg) is emerging, further complicating the dynamical landscape [10].

The congestion in GEO is amplified by the fact that many low-inclination objects exhibit limited orbital evolution over century-scale timespans in numerical simulations [11]. This orbital stability highlights the need for precise knowledge of each object's location and rotation state, as changes in attitude dynamics may serve as early indicators of fragmentation or collision events.

While end-of-life (EOL) disposal procedures, atmospheric re-entry strategies, and active debris removal (ADR) missions are effective in mitigating congestion in LEO, they are only partially applicable to GEO. Satellites nearing the end of their operational life are typically relocated to graveyard orbits, yet this mitigation alone may not ensure long-term orbital safety. One study [12] proposes a mission architecture able to relocate six derelict satellites to orbits above the current graveyard band, thereby reducing collision risks and preserving the GEO environment, yet implementation remains limited.

Understanding the rotational properties of inactive objects is critical for accurate modeling of external perturbations, thereby improving the reliability of long-term orbit predictions [13,14]. Additionally, spin state monitoring can indicate collision or fragmentation events, as well as the impact of space environment on satellites [15] and on their long-term attitude dynamics [9]. Accurate knowledge of spin state is particularly relevant for ADR, in-orbit life extension, and rendezvous operations, where attitude dynamics directly influence capture and docking strategies [16].

Photometric methods, especially lightcurve inversion, provide valuable insights into object shape, spin axis, and attitude [17,18]. Although it is a powerful method to infer all these parameters, many of the studies are only theoretical, based on simulated data. Even for simulated observations under realistic conditions, accurately reconstructing an object's true shape is challenging when using a single lightcurve, particularly in the presence of observational noise [19]. Furthermore, as the object's geometry becomes more complex, the simulations fail to recover the correct shape, even when more data is added into the analysis [19].

Previous photometric studies employed optical observations of artificial satellites using large aperture telescopes, equipped with CCD sensors, providing high resolution and small field-of-view (FOV). This configuration often implies slow download rate, thus decreasing the cadence of the images and leading to under-sampled data, particularly for fast-rotating objects.

A photometric study of few inactive GEO satellites over a one-month period [13] identified predominant rotators among the targets, with distinct lightcurve feature, despite similar construction and orbital characteristics. The authors emphasize the need for long-term and continuous monitoring in order to track the dynamical evolution. They have also addressed the limitations of the observational setup, where large aperture telescopes combined with CCD cameras, even when binned images were used, have a low temporal resolution, which undersample the lightcurves of rapid rotators.

The Astronomical Institute of the University of Bern (AIUB) built a decade-long lightcurve database, for over 400 objects across all orbital regimes [9]. Using a 1 m telescope equipped with a CCD camera, they reduced the readout time to 1 s by acquiring sub-frames. Both parametric and non-parametric analyses revealed that, while most targets were rotators with periods ranging from sub-second to 15 min, several stable objects with no detectable periodicity have also been identified.

Similar to previous studies, a long-term photometric monitoring of four inactive GEO satellites revealed spin periods ranging from 100 to

1 500 s [20]. Quasi-periodic spin period variations have been identified among satellites, attributed primarily to solar radiation pressure. The variation timescale for one individual satellite was greater than 520 days, exceeding the duration of the study, thus highlighting the importance of continuous long-term monitoring of same RSO. The same authors analyzed nearly four years of photometric data for 11 satellites, revealing both short term period variations, with timescales ranging from 200 days to over one year, and a long term evolution [21]. They emphasized the relevance of this data for dynamical modeling of Solar Radiation Pressure (SRP) torques, underscoring the necessity of long term monitoring of both previously studied satellites and newly selected ones.

An additional optical survey spanning several weeks confirms the presence of tumbling inactive satellites in GEO region [16]. Using large-aperture telescope and a CCD sensor for data acquisition, together with several tools for frequency analysis, the attitude motion of retired satellites was characterized. All observed objects exhibited tumbling motions, with the majority displaying angular velocities below $1^\circ/\text{s}$.

The first estimation of the spin axis orientation of an inactive box-wing GEO satellite revealed evidence of spin axis precession on a sidereal-year timescale [22]. However, the authors mentioned that equatorial coordinates of the spin axis have large uncertainties that could probably be caused by the sampling rate, thus addressing the need for higher cadence images and complementary observations to reduce orientation uncertainties.

Papusev et al. [15] conducted a photometric monitoring of uncontrolled GEO satellites between 1995 and 2003, determining that the rotation periods for formal three-axis stabilized spacecraft range from 20–430 s [15]. Based on long-term observations, three classes of inactive satellites were identified, according to their rotation period evolution: (1) gradual multi-year trends, (2) short-term spasmodic variations, and (3) brief anomalous increases over several days.

This paper introduces APSIS, an automated optical ground-based system designed for long-term, high-cadence photometric monitoring of satellites in the GEO and MEO regimes. While prior studies have demonstrated lightcurve-based rotation estimation for individual objects, the field currently lacks an operational, end-to-end framework capable of autonomous data acquisition, calibrated photometric reduction, and near real-time delivery of precise lightcurves for subsequent rotational and shape modeling. Although the pipeline supports both MEO and GEO targets, in this work we focus on inactive satellites in GEO, as their orbital geometry allows continuous visibility from a single observatory and multi-epoch monitoring, conditions required to resolve long-term variations in rotation period. In contrast to most existing optical systems that employ CCD detectors [13,16,23], our data are acquired using a sCMOS sensor which offers lower read noise and higher frame rates, features particularly suited to resolving the rapid photometric variability characteristic of artificial satellites. APSIS integrates real-time data acquisition, automated processing, and continuous production of calibrated lightcurves and astrometry with a lightcurve inversion pipeline based on a physically motivated, first-order model of the satellite, which we demonstrate to be capable of recovering the attitude and the principal physical characteristics of the object. The remainder of the paper is organized as follows: Section 2 details the system architecture, data acquisition strategy, astrometric and photometric reduction pipeline, and photometric analysis of the time series; Section 3 outlines the results of the rotational and inversion modeling framework validated on an inactive GEO satellite; and Section 4 summarizes the key findings and future work.

2. Observation setup and data acquisition

2.1. Berthelot Observatory

The data analyzed in this study was acquired using the infrastructure at the Berthelot Observatory (IAU code L54), which has been



Fig. 1. Berthelot observatory (left) and the dedicated SST system (right).

owned and operated by the Astronomical Institute of the Romanian Academy since its commissioning in 2018 [24]. The observatory is located in the commune of General Berthelot, Hunedoara County, Romania, and comprises three individual domes (Fig. 1), each dedicated to a specific domain within the field of Space Situational Awareness (SSA). The Berthelot-T05 telescope focuses on the study of Near-Earth Objects (NEOs), while the dual system composed of Berthelot-T04 and Haret-T02 is entirely dedicated to Space Surveillance and Tracking (SST) operations. Additionally, a recently commissioned solar observatory (currently in its testing phase) supports research in the Space Weather (SWE) domain [25].

The photometric study conducted in this work relies entirely on observational data obtained using the dual-telescope SST system (Berthelot-T04 and Haret-T02, Fig. 1), designed from the ground up for automated survey and tracking of artificial satellites across orbits ranging from LEO to GEO.

The Berthelot-T04 is a Ritchey-Chrétien telescope with a 0.4 m aperture and an $f/8$ focal ratio, configured for precision tracking of selected targets. Coupled with an Andor Marana 4.2B-11 camera (2048×2048 px; 32 mm diagonal) this configuration delivers an image scale of $0.8'' \text{ px}^{-1}$ for a $0.45^\circ \times 0.45^\circ$ field of view. Installed in 2021 on the same 10Micron GM4000 HPS II equatorial mount, Haret-T02 is a Riccardi-Honders telescope with a 0.20 m aperture and an $f/3$ focal ratio, offering a wide $2.15^\circ \times 2.15^\circ$ field of view. It employs an identical Andor Marana 4.2B-11 detector, providing an image scale of $3.78'' \text{ px}^{-1}$. The Andor Marana is a high quantum efficiency (95%), fast read-out time (0.3 s), low read-out noise ($1.6 e^-$) sCMOS camera for astronomy and physical sciences. Due to its low read noise, it can achieve higher signal-to-noise ratios than even high-end CCD cameras for short exposure times (up to a few seconds, before dark current becomes the dominant noise source), making it particularly well-suited for high-cadence photometric monitoring of rapidly varying phenomena [26]. The image acquisition system is based on Andor Software Development Kit (SDK). All elementary FITS (Flexible Image Transport System) operations such as creation, opening, updating, and arithmetic processing are implemented using the ESO Common Pipeline Library — CPL [27].

This combination of a high-resolution narrow-field system and a wide-field survey instrument allows us to both conduct precision follow-up measurements of selected satellites and efficiently monitor

larger sky areas to detect and characterize multiple targets within a single field.

The tracking and survey system has been part of the European Union Space Surveillance and Tracking¹ consortium (EU-SST) since 2019 (Berthelot-T04) and 2022 (Haret-T02), providing operational data in accordance with the network schedule. Both telescopes have successfully completed all internal calibration campaigns, demonstrating stable sensor performance and high measurement accuracy, including time-bias stability and robust astrometric performance [28].

2.2. APSIS - Automated Photometric Survey of Inactive Satellites

In our study we developed APSIS, a fully automated pipeline for the photometric monitoring and characterization of inactive satellites. The procedure is readily adaptable to different telescope-camera configurations, as demonstrated internally with both the Berthelot and Haret telescopes. APSIS provides real-time photometry by acquiring satellites and space debris images, processing the data, and generating lightcurves and associated rotational periods in near real time. These results are subsequently archived in an internal database for long-term monitoring of rotational behavior. Our approach prioritizes high temporal resolution, long-term photometric monitoring of selected inactive GEO objects, capturing them across multiple configurations to enable a comprehensive characterization of their spin state. We target the MEO and GEO regions, with a focus on space debris and inactive satellites that pose a potential risk to operational spacecraft.

APSYS was designed to operate in two complementary modes. In survey mode, the wide field telescope scans the GEO belt to provide short-timescale lightcurves for a large set of satellites and space debris while in the tracking mode, the narrow field telescope targets follow-up objects for long duration photometric monitoring of single targets.

Since 2023, we have been performing systematic GEO-belt survey observations in a continuous scanning mode to identify and classify inactive satellites by analyzing their photometric characteristics. The survey covers the GEO orbital arc visible from Berthelot Observatory. For a minimum target elevation of 14° , this corresponds to the orbital longitude interval $[80^\circ \text{E} : 30^\circ \text{W}]$. The survey was extended to include $\pm 2^\circ$ in declination around the GEO belt ($i = 0^\circ$) to account for

¹ <https://www.eusst.eu>

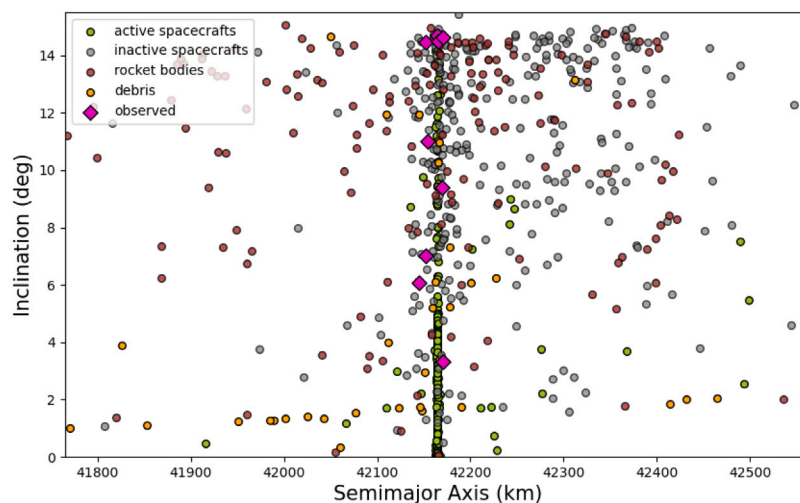


Fig. 2. Semi-major axis vs. inclination plot of observed objects against all RSO in geosynchronous protected region.
Data source: www.space-track.org.

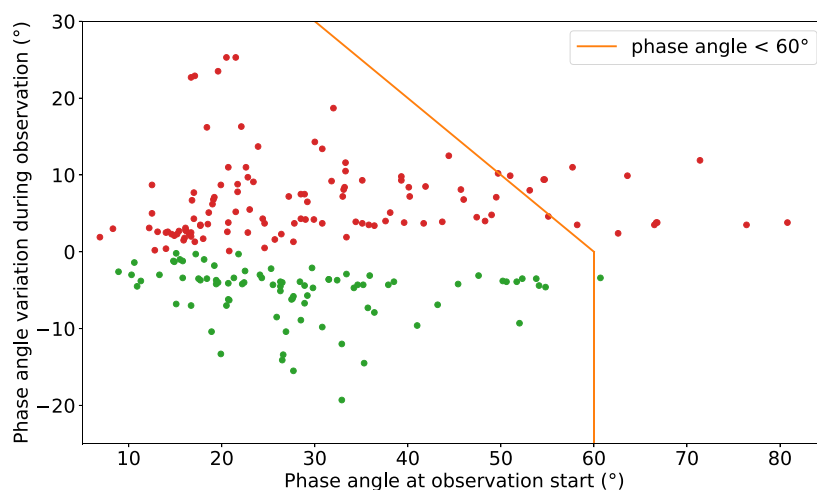


Fig. 3. Phase angle variation (increasing — red; decreasing — green) during photometric observations of long term monitored satellites. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

inclination drift and to increase the probability of detecting inactive, non-station-keeping objects. For each observed field, 30 consecutive images with 1 s exposure time (T_{exp}) are obtained. After completing the sequence, the telescope automatically slews to the next field to continue the survey. This strategy was found effective for surveying a large number of objects in a short time frame. The 30 s observing window was sufficient to capture the full rotation period for most targets, or at least to reveal significant brightness variations in the case of slow rotators.

Once detected and visually confirmed, the objects with a well-determined rotation period were scheduled for targeted follow-up observations.

For the system setup and performance evaluation, an initial list of candidate objects (Fig. 2) was compiled by filtering the SATCAT² catalog for entries located within the GEO protected region. The Inter-Agency Space Debris Coordination Committee (IADC) defines this region as the shell within $\pm 15^\circ$ in inclination and ± 200 km in altitude from the geostationary orbit [29].

Observations were scheduled according to the following criteria: a minimum target elevation of 14° ; a phase angle $\leq 60^\circ$ with the target

fully illuminated by the Sun for at least 30 min, ensuring no shadow entry during the run and a minimum angular separation of 10° from both the full Moon and the Galactic plane, in order to reduce sky background and avoid star-crowded fields that could contaminate the photometric measurements.

Visibility windows satisfying all criteria were computed in advance for each object and subsequently incorporated into the overall observation plan. Priority was given to satellites exiting the visibility window within one hour from the start of observations, and to those with an increasing phase angle during the observing session, as this corresponds to less favorable illumination and observing conditions (Fig. 3). Using the daily updated SATCAT Two Line Element (TLE) catalogue, APSIS computes, for each target, the visibility windows over a 48-hour interval, the topocentric right ascension (RA) and declination (Dec) and the phase angle. Internally, APSIS uses NOVAS C library [30] to perform high-precision astrometric calculations and SGP4/SDP4 orbit propagation models [31,32] for accurate prediction of satellite positions.

Fig. 3 illustrates the variation in phase angle during photometric runs as a function of the initial phase angle. The sample is nearly evenly split between cases of increasing (red) and decreasing (green) phase angle. A substantial fraction of the observations remain entirely below 60° , ensuring favorable illumination conditions for reliable

² www.space-track.org

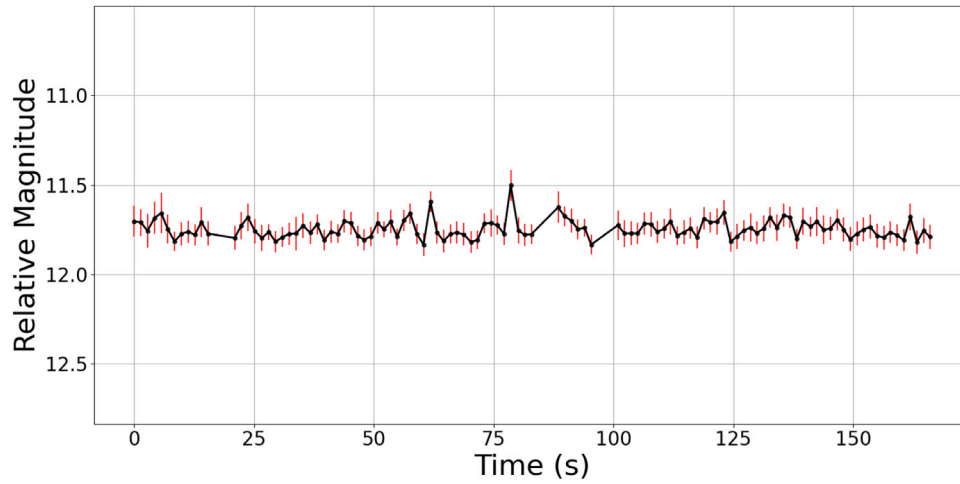
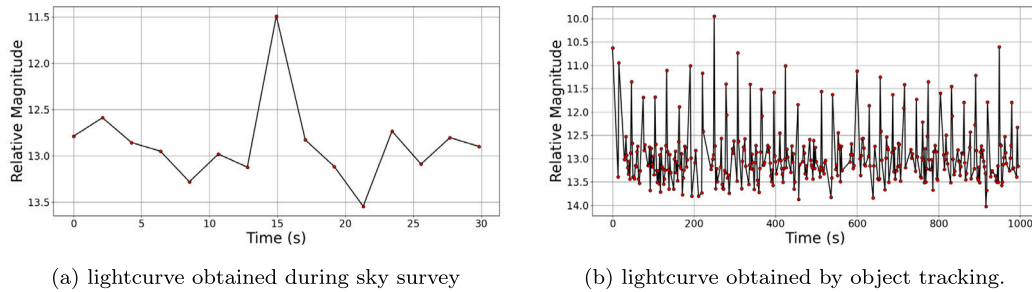


Fig. 4. Active, attitude stabilized satellite (Turksat 6A, 2024-127A) lightcurve obtained during photometric surveys.



(a) lightcurve obtained during sky survey

(b) lightcurve obtained by object tracking.

Fig. 5. Lightcurve of 1991-014A obtained during survey and follow-up using the telescope in tracking mode.

photometric analysis. The wide range of phase angles covered in our observations provides valuable geometric constraints for the characterization of the tracked objects, as variations in illumination and viewing geometry help disentangle shape, attitude, and surface reflectance effects in the lightcurves. We note that APSIS automatically identifies, tracks, and generates lightcurves for any objects drifting through, or already present in the field of view of our target, including active, attitude-stabilized satellites (Fig. 4). These cases serve as a baseline for evaluating the stability of the photometric solution throughout the night.

A representative case of uncontrolled object detection is presented in Fig. 5, where the raw lightcurve of the inactive satellite 1991-014A, obtained during the survey, exhibits brightness variations with an amplitude of 1 magnitude over a 30-second observation window. This amplitude met the minimum threshold for further analysis, which requires unprocessed lightcurves to display photometric variations of at least 1 magnitude. The initial detection was confirmed by follow-up dedicated observations, during which the same satellite was tracked for 16 min (Fig. 5b).

Drift rate and revisit time were also used as selection criteria to ensure sufficient temporal coverage for characterizing attitude evolution. Fig. 6 illustrates the drift rate of selected satellites as a function of its altitude offset from the nominal GEO altitude, together with the corresponding revisit time defined as the number of years required for a satellite to return to the initial orbital longitude. Because the observatory monitors a longitudinal arc of $\approx 110^\circ$, satellites re-enter the observable region after drifting $\approx 250^\circ$ of longitude, shortening the effective revisit time relative to a full 360° rotation.

Additionally, the satellite's orbital altitude influences the optimal observation time. To maximize observational coverage, objects are tracked as soon as they enter the longitudinal arc accessible

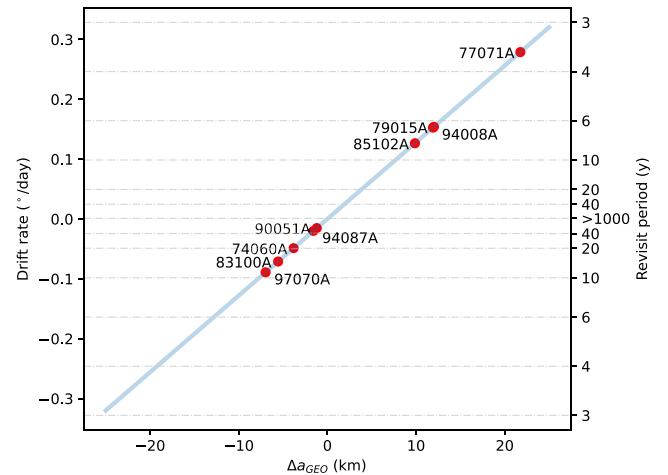


Fig. 6. Drift rate and revisiting time of studied satellites.

from the Berthelot Observatory, that is, when eastward-drifting (sub-synchronous) and westward-drifting (supra-synchronous) satellites reach orbital longitudes of 30°W and, respectively 110°E . For optimal phase angles, the former are best observed late at night, while the latter are best observed shortly after sunset. This strategy ensures observations at an optimal phase angle for objects which, although drifting, will remain within the observatory's tracking capabilities for extended periods. Based on the criteria discussed above, the system compiles at the start of each night a list of observable targets, previously identified as variable during the survey phase, and submits it to the telescope scheduler for execution phase.

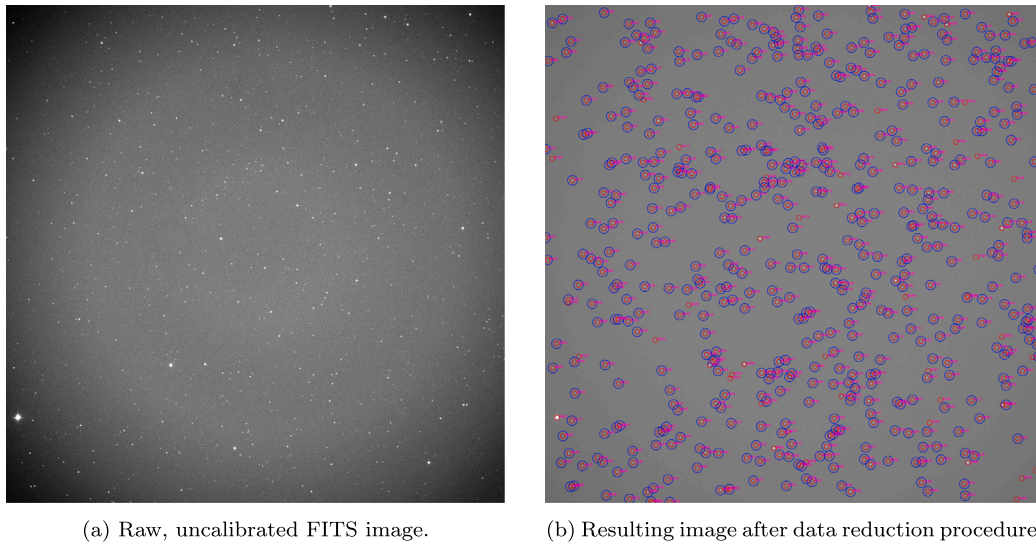


Fig. 7. Example frame acquired with the Haret-T02 telescope. Date: 2025-04-09T22:17:06.971 UTC/Center RA = 248.083°, DEC = −7.116° / $T_{\text{exp}} = 1.5$ s. Raw image (a) and calibrated image (b) showing the stars used for plate solving (concentric blue and red circles) and objects flagged as possible satellites (red circles only). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

APSIS integrates custom-developed tools written in C and Python to handle the entire data flow, from image acquisition and reduction, to lightcurve generation and period extraction. The pipeline presented in Fig. 8 performs automated image calibration, reduction, and photometric analysis of the observational data. At the beginning of each night, both telescopes automatically acquire calibration images (bias, dark, and flat frames), which are subsequently used in standard calibration procedures to remove electronic noise and thermal signal contributions and to compensate for non-uniform illumination and detector response. These calibration frames are refreshed nightly to monitor and correct for any changes in detector performance over time.

The photometric follow-up observations were always performed in tracking mode, with tracking rates computed in real time from the most recent TLEs and continuously applied to update the telescope mount during the run. Source detection was performed using the Source Extraction and Photometry (SEP) C library [33], which provides the core algorithms of SExtractor [34] as a library of stand-alone functions, replicating most of its functionalities. Aperture photometry [35] was performed using a standard circular aperture of $3 \times \text{FWHM}$ (approximately $6''$) for the target, and a $15''$ aperture for the trailed stars. Astrometric reduction used a subset ($G_{\text{mag}} < 15$) of GAIA DR2 catalog, with the right ascension and declination of reference stars propagated to the observation epoch using the virtual_place routine [36,37] implemented in the NOVAS-3.1 C library. The propagated positions were projected onto the tangent plane and matched to extracted sources positions using the triangles method [38]. Using the image center coordinates from the FITS header, together with a per-telescope configuration file containing the image scale, field-of-view dimensions, and limiting magnitude, this procedure proved robust and capable of astrometrically solving wide-field images with up to 10 000 stars in only a few seconds. The World Coordinates System (WCS) solution is computed using a linear transformation for the tracking telescope, while a cubic polynomial transformation is applied for the wide-field telescope to account for optical distortions across the larger field of view (Fig. 7). The global RMS of the astrometric solution is typically below $0.5''$ for the tracking telescope and $0.7''$ for the survey telescope. Although precise timing is less critical for photometric observations than for astrometry, both systems, NTP-disciplined, demonstrated during the EU-SST periodic calibration campaigns a stable timebase with a time-bias drift measured from optical observations of Galileo satellites of less than 10 ms year-to-year. This stability, while exceeding the requirements for photometry, ensures accurate correlation between

lightcurves and ephemerides, supports precise rotational phase determination in long-term monitoring, and facilitates cross-comparison of results with other instruments and observation campaigns.

The detection process operates on sequential triplets of plate-solved frames by identifying objects whose apparent motion in right ascension and declination remains linear across the three exposures. APSIS is currently tuned to detect objects in the MEO to GEO regime, with an apparent angular velocity acceptance range from $5'' \text{ s}^{-1}$ to $100'' \text{ s}^{-1}$. Because the object motion is extracted directly in celestial coordinates, the detection algorithm is agnostic to the target morphology (point-like or trailed) and, consequently, to the observing mode (survey or tracking). For each satellite detected in all three consecutive images, a Tracking Data Message (TDM) file is produced in compliance with the CCSDS standard [39]. The TDM contains astrometric positions expressed in the EME2000 (J2000) equatorial frame, corrected for annual aberration.

The satellite's apparent magnitude is determined via linear regression between the instrumental magnitudes and the catalog magnitudes of reference stars (Fig. 9), yielding a calibrated magnitude in the Gaia G -band system, which is recorded in the TDM file. Outliers, typically extended sources (galaxies) or blended object, are iteratively rejected using a 3σ kappa-sigma clipping algorithm. For each frame, the final linear regression provides a global RMS of the fit representing the photometric calibration uncertainty. The final magnitude uncertainty is calculated as $\sigma_{\text{mag}} = \sqrt{\sigma_{\text{SEP}}^2 + \sigma_{\text{global}}^2}$ where σ_{SEP} is the SEP-derived photometric error of the target and σ_{global} is defined above. This method also enables the estimation of the telescope's limiting magnitude under the current observational conditions, defined as the faintest detectable source with reliable signal-to-noise ratio. In our configuration, the limiting magnitude for a 1 s exposure time typically reaches ≈ 14 mag for the survey telescope and ≈ 15 for the tracking telescope.

Once generated, the TDM files are automatically cross-matched with SATCAT TLEs of satellites in the MEO to GEO regime, propagated to the observation epoch using the SGP4/SDP4 algorithms. A detection threshold of $60''$ from the propagated ephemeris is applied for TLEs with an age of less than two days, increasing to $200''$ for older TLEs. These thresholds account for the degradation of orbital predictions with increasing TLE age, which leads to larger positional uncertainties for older elements. Finally, APSIS compiles, for each successfully correlated object, a complete track consisting of all its detections during the observation session. The RMS of the astrometric residuals in right ascension

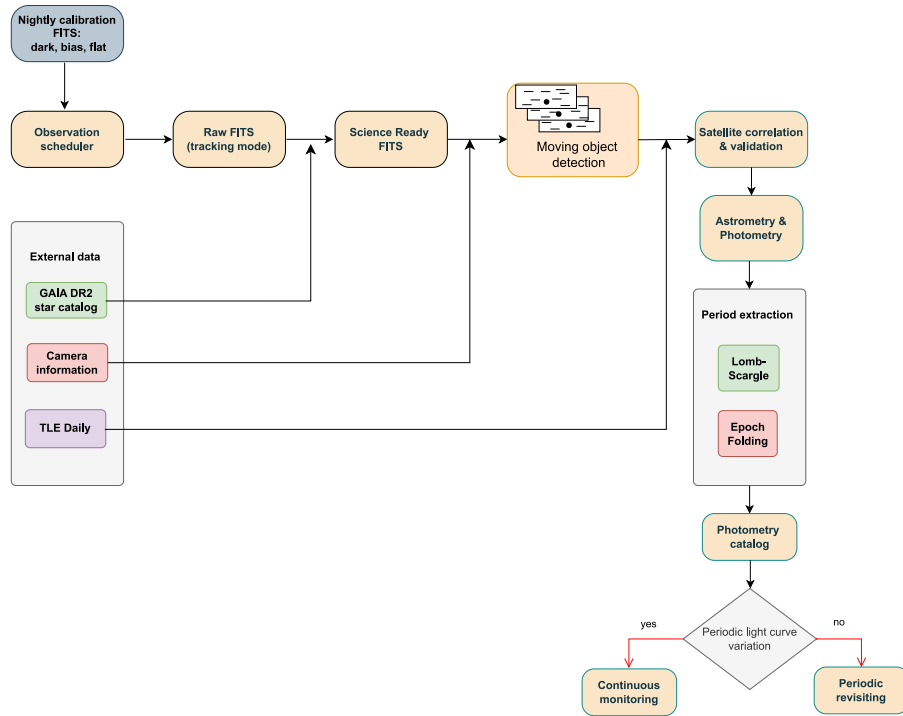


Fig. 8. Operational diagram of the APSIS automated procedure.

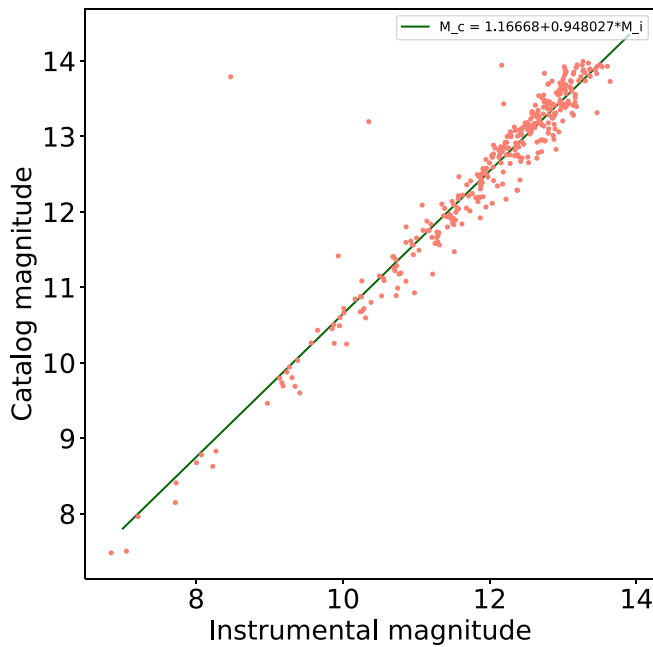


Fig. 9. Instrumental vs. catalog magnitude for a frame.

and declination, with respect to the most recent TLE, is evaluated to ensure that the object is consistently detected across all frames and that no interlopers contaminate the resulting lightcurves. All accepted tracks are archived in the APSIS database to support long-term monitoring of each object.

2.3. Lomb–Scargle (LS) periodogram

Rotational periods were derived from detrended photometric time series using the Generalized Lomb–Scargle (GLS) periodogram [40],

which provides a least-squares solution for sinusoidal fitting under irregular temporal sampling. Up to three harmonic terms were included in the model, which was found to provide adequate sensitivity to non-sinusoidal lightcurve shapes while avoiding overfitting and spurious peaks in sparsely sampled data [41]. The scanned frequency range was 5×10^{-4} Hz to 1 Hz, corresponding to rotational periods between 1 s and 2000 s range that includes typical values for inactive or tumbling geostationary satellites [20]. The frequency corresponding to the maximum periodogram power was taken as the initial estimate. A local Gaussian fit to the spectral peak was then used to refine the frequency, yielding the final rotational period. The Gaussian width provided the associated 1σ uncertainty.

A known limitation of LS is the tendency to return a frequency twice the true rotation frequency when the lightcurve exhibits a double-peaked symmetry. This situation is common for objects with two similar reflective facets separated by 180° , producing nearly identical brightness maxima each half-rotation. In such cases, the periodogram power at $2f_{\text{true}}$ can exceed that at f_{true} , leading to an apparent rotation period of $P_{\text{obs}} \approx P_{\text{true}}/2$ [41]. In one such case (Fig. 10), the strongest peak in the Lomb–Scargle periodogram occurs at a period of 6.1 s, while the true period is 12.2 s. This illustrates harmonic dominance in double-peaked lightcurves, where the first harmonic can exceed the fundamental. To mitigate this, all candidate periods are visually verified by folding the lightcurve at both P_{obs} and $2P_{\text{obs}}$.

The custom Python implementation generates, for each processed lightcurve, a set of four diagnostic plots that support the determination and verification of the rotation period: (1) the raw lightcurve (magnitude vs. time), (2) the power spectrum with the identified frequency peak, (3) a Gaussian fit to the peak for refined period estimation and uncertainty, and (4) the phase-folded lightcurve constructed using the final period. An example output for the satellite 1977-071A (Raduga 3), one of our longest-monitored targets and representative of the high-cadence, multi-epoch photometry obtained with APSIS, is shown in Fig. 11.

2.4. Epoch folding

A secondary, independent method based on *epoch folding* using a phase dispersion minimization algorithm [42] was applied to validate

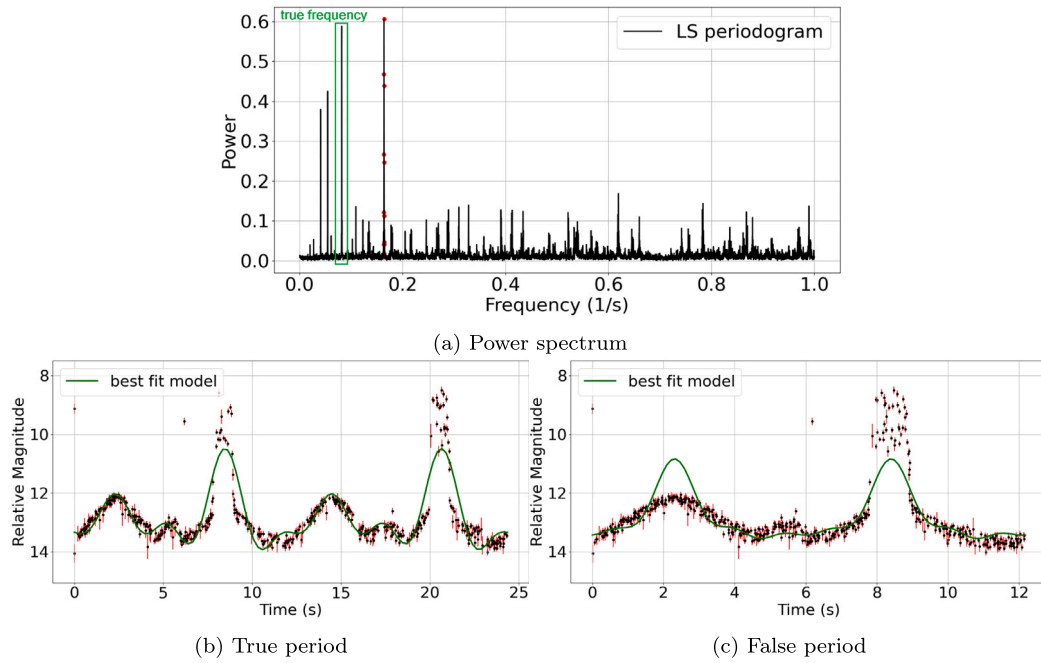


Fig. 10. Lomb–Scargle periodogram (a) showing a dominant peak at $2f_{\text{true}}$ corresponding to a period of 6.1 s. The true period is 12.2 s (b). RADUGA 3 satellite (1977-071A).

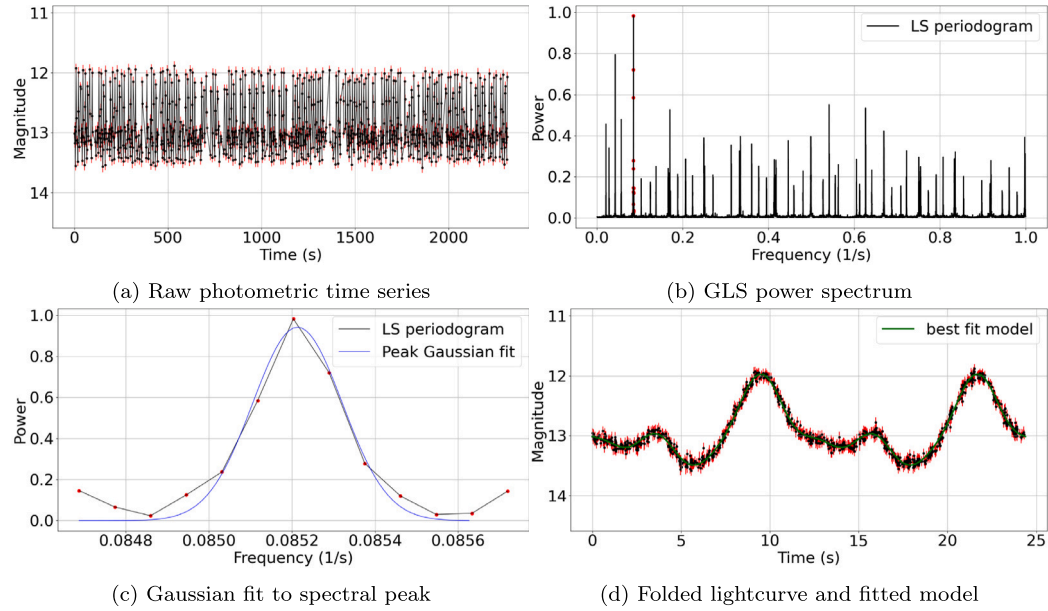


Fig. 11. Photometric period analysis for GEO object 1977-071A. (a) Raw lightcurve showing magnitude variation over time. (b) Generalized Lomb–Scargle power spectrum with the dominant frequency. (c) Gaussian model fitted to the main peak to refine the frequency and estimate the 1σ uncertainty. (d) Phase-folded lightcurve based on the final period solution, overplotted with the fitted multi-harmonic model. Data obtained on 2023-07-15.

and, if necessary, refine the candidate periods derived from the GLS analysis. In this approach, the lightcurve is phase-folded over a dense grid of trial periods in the range 2–200 s, with a step size of 0.001 s. For each trial period, the data are binned in phase and the photometric dispersion of the folded lightcurve is computed. The trial period yielding the minimum dispersion is adopted as the best-fit solution (Fig. 12). This method is less sensitive to waveform shape and sampling aliases than GLS, making it particularly effective for highly non-sinusoidal or irregular lightcurves. Final validation was performed through visual inspection of the folded lightcurves to ensure the period provided a consistent and coherent modulation pattern across all phases.

3. Results

APSYS is capable of monitoring a large number of objects in the MEO to GEO regime, continuously updating their rotation periods and providing accurate, up-to-date input for dynamical modeling tools. This capability enables the construction of long-term, homogeneous time series for a wide range of targets.

To investigate long-term rotational dynamics, we monitored the temporal evolution of the rotation period for a representative sample of inactive nearly geosynchronous satellites (Fig. 2). In contrast to previous findings [13], no tumbling objects were identified among our targets. The resulting time series revealed, for most objects, a clear

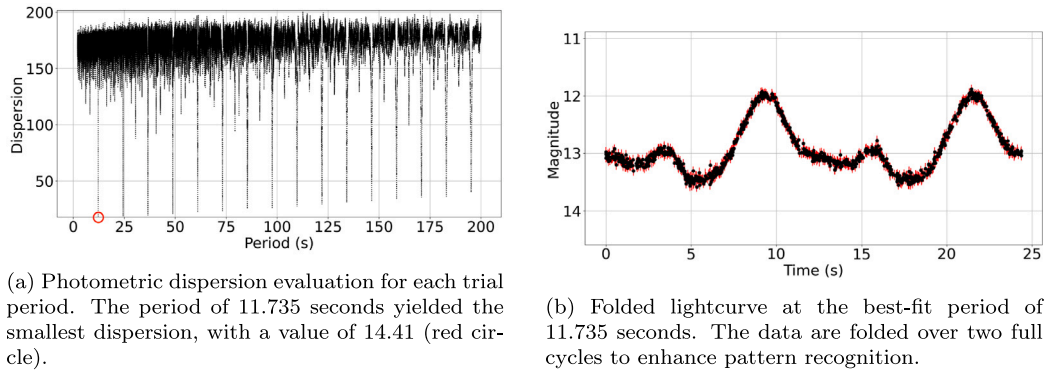


Fig. 12. Epoch folding analysis of GEO object 1977-071A (Raduga/KAUR-3 class).

seasonal modulation superimposed on a secular trend. In the majority of cases, the long-term trend corresponded to a net acceleration in the spin rate, while for 1994-087A and, to a lesser extent for 1977-071A, a gradual deceleration was observed (Fig. 13, Appendix A). Changes in the rotation periods of GEO satellites have been reported previously and attributed to internal mechanisms or to weak external forces acting on the satellites [15]. Cyclic variations of the rotation period have been attributed to torques induced by solar radiation pressure acting on their large-area solar panels [20]. Periodic oscillations in the rotation period have also been reported for decommissioned GLONASS satellites and attributed to solar radiation pressure acting on canted solar panels [43], while MEO rocket bodies and space debris show no detectable variation in rotation period [44]. Consistent with these findings, our continuous monitoring confirms this effect across multiple objects and epochs and provides a systematic characterization within our dataset.

3.1. Solar radiation pressure and period evolution

To quantitatively characterize both components, the periodic (seasonal) modulation and the linear (secular) drift, we modeled the measured periods using:

$$P(t) = P_0 + A \cos(\omega t + \phi) + B(t - t_0) \quad (1)$$

where P_0 (the initial period) and B describe the secular trend of the data while A , ω , and ϕ characterize the amplitude, period, and phase of the variation. Table 1 summarizes the parameters obtained from fitting the time evolution of the measured rotation periods for our set of satellites. For each object we list the best-fit period of the modulation, the amplitude of the seasonal term, and the slope of the long-term drift. The fitted periods cluster around one year, a timescale that strongly suggests a forcing mechanism related to the annual variation of the Sun's apparent position (declination). We interpret this near-annual periodicity as being generated by the annual variation of solar radiation pressure torques, which depend on the geometry of the Sun-object orientation. This result is clearly seen in the case of high amplitude, long baseline, higher quality data (1977-071A and 1990-051A) where annual variation of the rotational period dominates the overall trend of the data.

To further investigate the link between the seasonal modulation and the solar forcing, we computed the phase lag between the detrended period variations and the Sun's declination. After removing the long-term linear trend from the period data, the remaining periodic component was cross-correlated with the solar declination time series. The position of the peak in the cross-correlation function provides an estimate of the time delay between the changes in the rotation period and in the Sun declination. This procedure allows a robust determination of the time

Table 1

Lag, period, and slope for selected objects.

Satellite	Season Lag (days)	Period (days)	Seasonal amplitude (s)	Slope (ms/day)
1974-060A	38.12 ± 1.86	347.3 ± 12.8	0.020	-0.02
1977-071A	170.92 ± 2.04	360.4 ± 1.40	0.610	0.92
1979-015A	-8.04 ± 1.45	350.9 ± 43.2	1.070	-25.64
1983-100A	0.34 ± 1.49	348.7 ± 3.00	0.039	-0.56
1985-102A	131.16 ± 1.98	349.4 ± 23.6	0.060	-0.11
1990-051A	-181.15 ± 2.11	361.5 ± 1.50	0.479	-1.67
1994-008A	92.61 ± 1.98	364.2 ± 2.40	0.080	-1.16
1994-087A	-133.94 ± 1.97	342.9 ± 13.6	3.360	43.55
1997-070A	-143.35 ± 1.99	328.4 ± 4.20	0.045	-1.11

shift, even when the data are unevenly sampled. The lag quantifies how the response of the rotation period follows the annual cycle of solar declination, and the results for each object are reported in Table 1. The detrended periods and the Sun's declination are presented in Fig. 14.

The cross-correlation analysis shows that the phase of the seasonal modulation varies among objects. For example, in the case of 1977-071A and 1990-051A the lag is about 180 days, with extrema of the rotation period occurring close to the solstices and aligned with the extrema of the Sun's declination. This corresponds to a nearly perfect anti-correlation between the rotation period variations and the annual solar forcing. In contrast, for 1994-008A we identify a lag of about 90 days, so that the rotation period extrema are found close to the equinoxes, when the Sun crosses the equatorial plane. These patterns demonstrate that the periodic changes are closely tied to the Sun illumination geometry, but with an inverted or delayed dynamical response to Sun position.

The most plausible candidates for this effect is Solar Radiation Pressure, which exerts a torque on asymmetrically shaped satellites when the incident sunlight does not pass through the center of mass. This behavior may arise from purely geometrical effects, for example if the dominant solar radiation pressure torque reverses sign between the Northern and Southern hemispheres due to the orientation of reflective surfaces, or it could reflect a delayed dynamical response of the satellite itself, driven by changes in its internal mass distribution (e.g., propellant sloshing) or the action of attitude control systems such as reaction wheels. Even for satellites whose reaction wheels have long been deactivated, spinning components or free-moving internal elements may produce a delayed response to external torques. Reaction wheel may even switch-on spontaneously under favorable solar panels illumination conditions [15] and can generate abrupt changes of the rotation period.

For most inactive objects, there is no reliable information on their internal structure: mass distribution, remaining propellant in tanks,

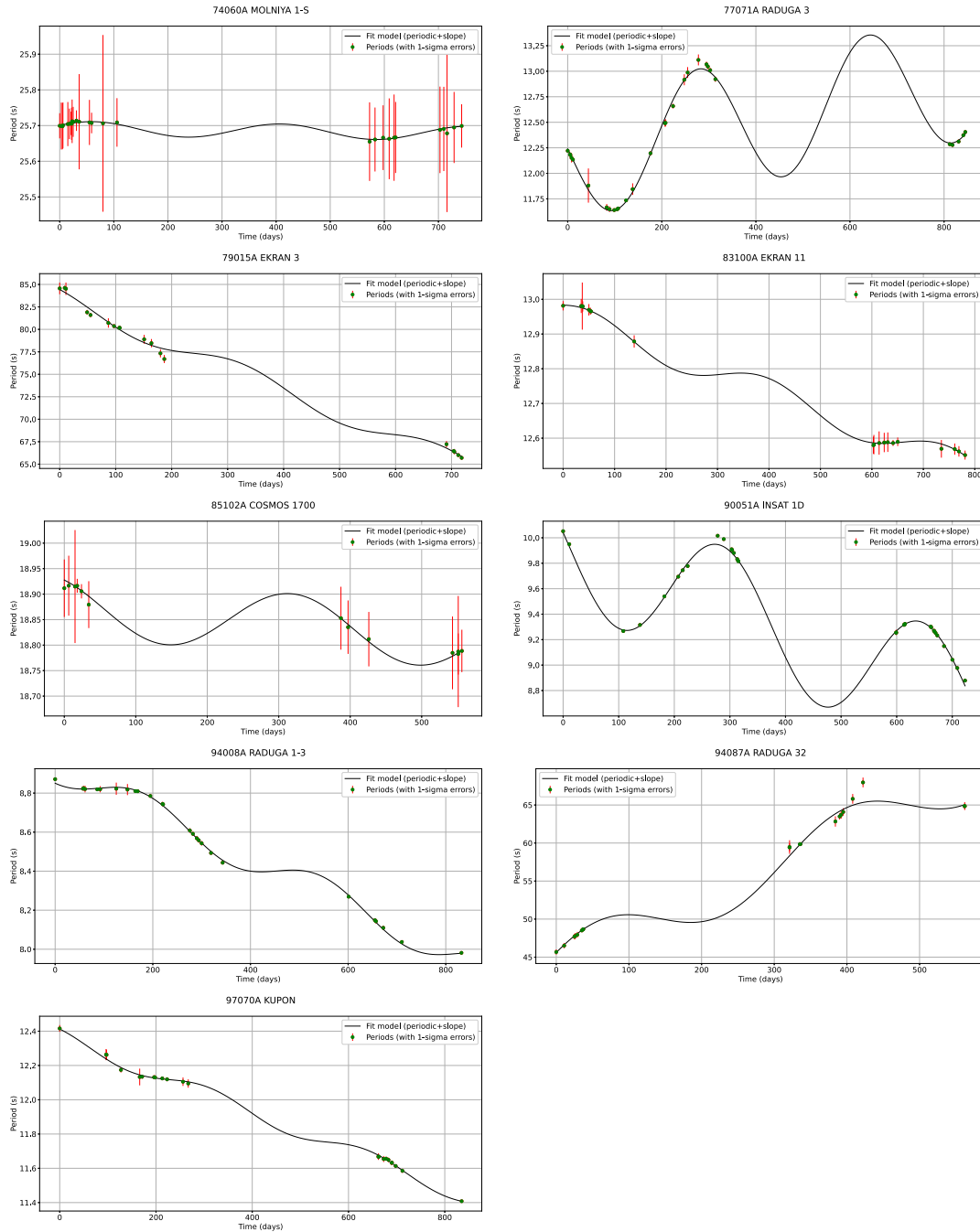


Fig. 13. Rotation period evolution for the nine analyzed satellites.

reaction wheels size and location or final orientation of the solar panel. All these elements can modify the satellite response to external torques. These internal effects can introduce complex and sometimes delayed responses that are difficult to model without direct knowledge of the object.

In contrast, the geometrical forcing by the Sun defined by its apparent motion relative to the satellite rotation axis is predictable. A critical part of the analysis is therefore to disentangle the contribution of this external, predictable driver from the unknown internal effects. By explicitly accounting for the solar geometry and its periodicity, we can assess how much of the observed modulation is explained by the Sun position relative to the satellite and isolate any residual behavior that must originate from the satellite internal dynamics.

3.2. Rotation-period evolution and spin-axis orientation

The period evolution can also be modeled as a response to solar radiation pressure torques, expressed as

$$P(t) = P_0 + A \cos \psi(t) + B(t - t_0), \quad (2)$$

where $\psi(t)$ is the angle between the satellite spin axis and the satellite-to-Sun vector. The amplitude A incorporates all physical scaling factors, such as effective area, reflectivity and torque lever arm.

If the spin axis is specified by equatorial coordinates $(\alpha_{\text{pole}}, \delta_{\text{pole}})$ and the solar position at time t by $(\alpha_{\odot}(t), \delta_{\odot}(t))$, then

$$\cos \psi(t) = \sin \delta_{\text{pole}} \sin \delta_{\odot}(t) + \cos \delta_{\text{pole}} \cos \delta_{\odot}(t) \cos (\alpha_{\odot}(t) - \alpha_{\text{pole}}) \quad (3)$$

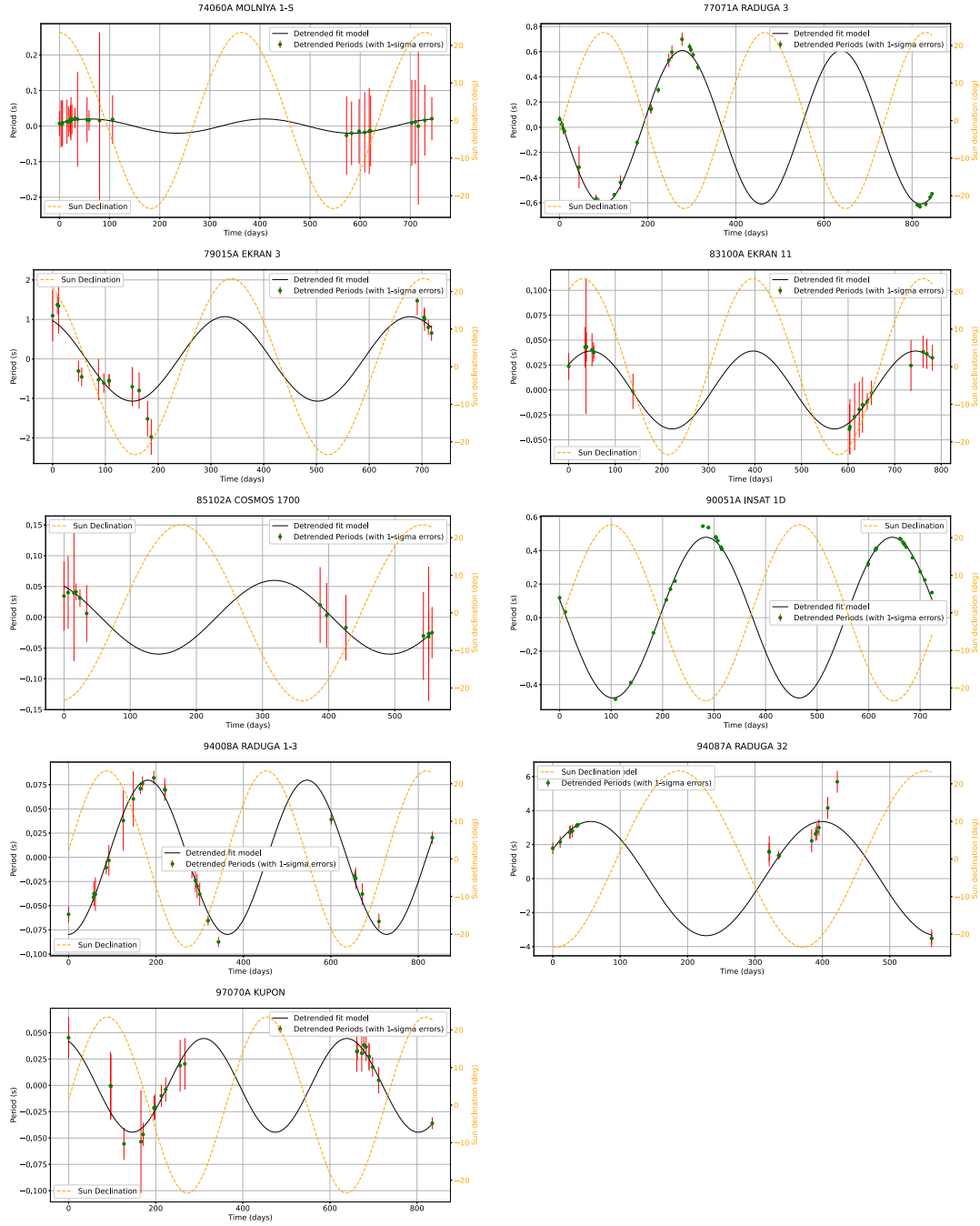


Fig. 14. Detrended rotation period and the Sun's declination.

To explore possible pole orientations, we performed a grid search over $\alpha_{\text{pole}} \in [0^\circ, 360^\circ]$ and $\delta_{\text{pole}} \in [-90^\circ, +90^\circ]$ with 0.5° steps. For each grid point we computed $\cos \psi(t)$ and refitted P_0 , A , and B by weighted least squares. The best-fit pole was defined as the minimum of

$$\chi^2 = \sum_i \left[\frac{P_i - P(t_i)}{\sigma_i} \right]^2. \quad (4)$$

where P_i and $P(t_i)$ are the measured and model-fitted rotation periods, respectively, and σ_i are the associated uncertainties.

Uncertainties were explored by generating 100 random $(\alpha_{\text{pole}}, \delta_{\text{pole}})$ clones within the 6σ confidence region, each individually refitted for P_0 , A , and B . The results are shown in Fig. 15 in the upper panels, where the detrended measurements are compared with the best-fit model (red) and its associated $\cos \psi(t)$ (orange), while gray curves illustrate clone

solutions. The lower panels display the χ^2 maps with best-fit poles (red diamond) and 6σ contours (white).

In most cases the χ^2 surfaces have narrow domains separated by 180° in right ascension, reflecting the intrinsic symmetry of Eq. (2). This means that many pole orientations provide statistically indistinguishable fits once amplitude and slope are re-optimized. As a consequence, the method cannot provide unique pole orientation, but rather families of statistically equivalent solutions. This approach therefore demonstrates consistency between SRP forcing and the observed periodicities, while highlighting the inherent degeneracy of such fits and the need for complementary constraints (e.g., lightcurve shape modeling) to resolve the true spin-axis geometry.

Our analysis up to this point has focused exclusively on the geometric configuration between the Sun and the satellite's rotation axis,

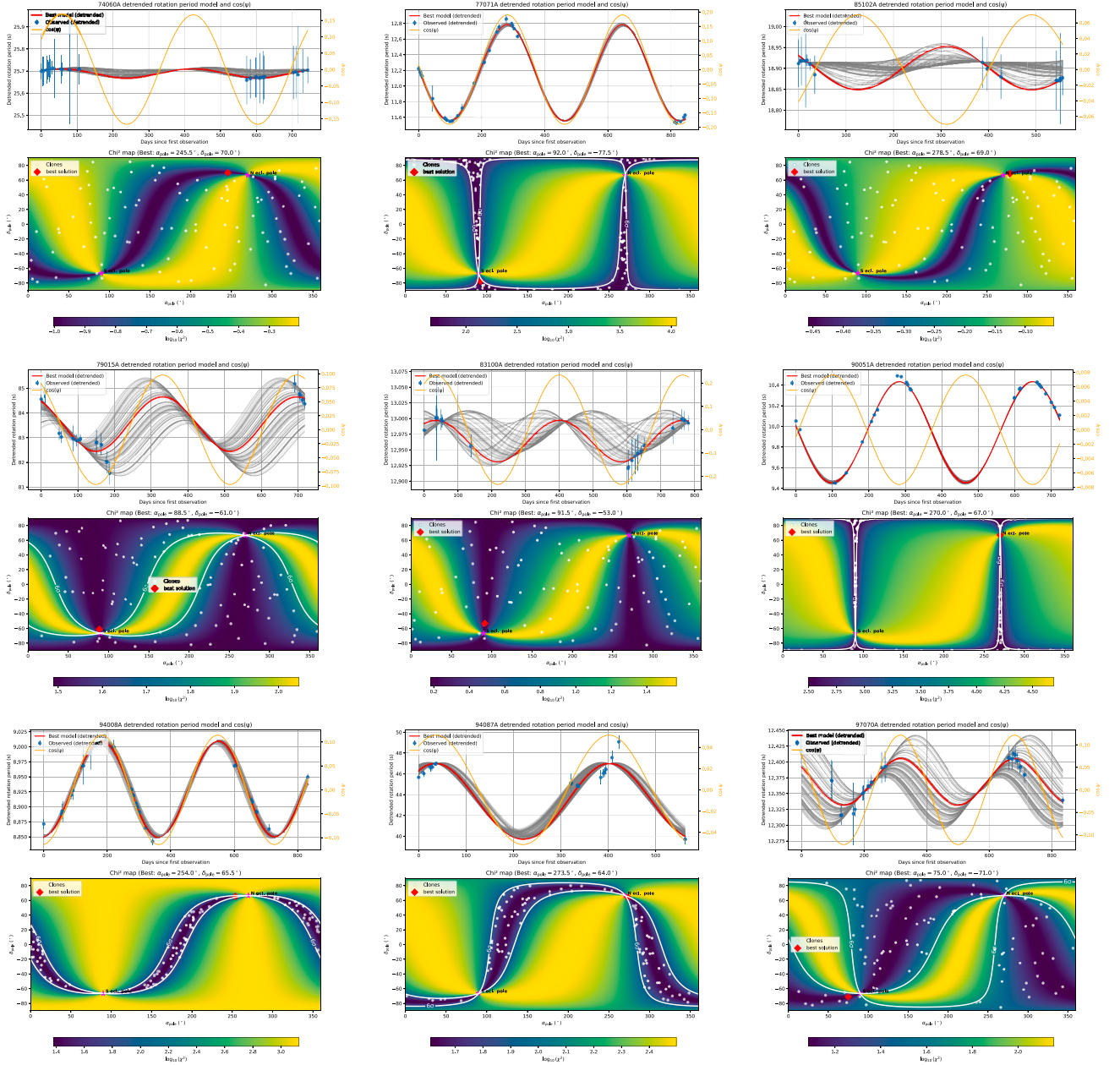


Fig. 15. Rotation period model and pole solution space for all our objects.

without incorporating the physical properties of the observed objects. APSIS aims to characterize artificial satellites using the smallest set of physically relevant parameters needed to describe their long-term photometric behavior. This approach does not rely on detailed shape or structural models, which are often unavailable or classified. Instead, the survey emphasizes modeling based on photometric data acquired automatically: rotation period and its changes and multi-epoch lightcurve analysis.

By focusing on key signatures (e.g., periods evolution, seasonal modulation, lightcurves shapes), APSIS enables the classification of inactive satellites into dynamical or morphological families, supports diagnostics of rotational stability, and provides predictive capabilities for future behavior, including spin evolution under solar torque. This minimal-parameter strategy ensures rapid characterization of a large population of objects, within a simplified, computationally feasible model that reflects real physical drivers of satellite rotation, such as SRP torque and body geometry.

3.3. Light curve inversion

Lightcurve inversion, originally developed for deriving the shapes and spin states of unresolved asteroids [45–48], is equally applicable to artificial satellites observed as point sources. In both cases, a lightcurve encodes information on the object’s geometry, reflective properties, and attitude evolution. This approach enables non-resolved, ground-based photometry to provide realistic estimates of spacecraft attitude and shape characteristics without the need for direct imaging. Lightcurve inversion has been applied to multi-site simulated photometric data to recover the basic polyhedral shape of a satellite [49], as well as to estimate the attitude of satellites modeled with complex geometries [50]. Given the complexity and computational cost of the problem, lightcurve inversion procedures have often approached shape and attitude estimation separately [51–53] or have required an initial knowledge of the attitude state [54,55]. Recently, particle swarm optimizer has been used to infer attitude and optical properties of RSO from simulated

space-based optical data [56,57] and from simulated ground-based observations [58].

For tackling high-volume, automated photometric survey data, we propose a simplified, physically motivated model of space objects. In this framework, the satellite is represented as a diffuse reflecting rectangular box (the bus) with six faces following the Lommel–Seeliger law [59,60], together with a single specular (glinting) panel fixed in the body frame, representing a solar array or another component (e.g., radiator, directional antenna) with a strong, highly anisotropic bidirectional reflectance distribution function (BRDF). Because the bus term represents all diffuse reflection, any diffuse contribution from the glinting panel is implicitly included in the bus flux, while the glint term describes only the geometry and peak amplitude of the specular component. A limitation of this model is the absence of self shadowing between different satellite components. In reality, structural elements such as solar arrays, radiators, and antenna booms can shadow parts of the bus or other panels depending on the Sun-satellite-observer geometry. This mutual shading can significantly alter the illuminated area and thus the instantaneous flux. In our simplified framework, all surfaces are assumed to be fully illuminated whenever their solar incidence angle is positive, which may overestimate the brightness during certain rotation phases or geometries with large occluding structures. While this model cannot reproduce the exact real-world geometry of a satellite, it captures the key features responsible for generating the observed lightcurve shapes.

In our model, the bus geometry is parameterized by the relative axis ratios b/a and c/a , where a , b , and c correspond to the dimensions along the body-frame x -, y -, and z -axes, respectively. The parameter A_{box} acts as a global diffuse flux scale factor, so the values of a , b , and c are interpreted only as relative dimensions rather than exact physical measurements. The bus z -axis is aligned with the satellite's rotation axis. The glinting panel is specified by its body-frame co-latitude θ_g and azimuth ϕ_g relative to the spin axis (Fig. 21). This decoupling of size and flux is a common approach in unresolved photometric modeling of spacecraft, originating from asteroid lightcurve inversion techniques, and allows shape parameters to be constrained independently of absolute scale [61].

The total modeled flux is given by the sum of the diffuse (bus) contribution and the specular (glint) contribution:

$$F_{\text{total}} = F_{\text{diffuse}} + F_{\text{glint}} \quad (5)$$

The diffuse flux is computed as the sum over all six bus faces, each following the Lommel–Seeliger law:

$$F_{\text{diffuse}} = \sum_{i=1}^6 A_{\text{box}} S_i \begin{cases} \frac{\mu_{\odot,i} \mu_{o,i}}{\mu_{\odot,i} + \mu_{o,i}}, & \mu_{\odot,i} > 0 \text{ and } \mu_{o,i} > 0, \\ 0, & \text{otherwise,} \end{cases} \quad (6)$$

where: A_{box} is the global diffuse flux scale factor, S_i is the relative area scale for face i ($S_i \in \{1, b/a, c/a\}$ depending on face orientation), $\mu_{\odot,i} = \hat{\mathbf{s}}_{\odot} \cdot \mathbf{n}_i$ and $\mu_{o,i} = \hat{\mathbf{e}} \cdot \mathbf{n}_i$ are the cosine of the incidence and of the emergence angle for face i , $\mathbf{n}_i$ is the unit normal of face i in the inertial frame at the given rotation phase and $\hat{\mathbf{s}}_{\odot}$ and $\hat{\mathbf{e}}$ are the unit vectors from the satellite to the Sun, and from the satellite to the observer respectively, computed from the ephemeris at the lightcurve epoch.

The glint flux is computed from a single panel with body-frame orientation defined by co-latitude θ_g and azimuth ϕ_g :

$$F_{\text{glint}} = \begin{cases} G \exp\left[-\frac{\psi^2}{2\sigma^2}\right], & \mu_{\odot,g} > 0 \text{ and } \mu_{o,g} > 0, \\ 0, & \text{otherwise,} \end{cases} \quad (7)$$

where: G and σ are the glint peak amplitude and angular width (degrees) and $\mu_{\odot,g}$ and $\mu_{o,g}$ are defined as in the case of the bus considering $\mathbf{n}_g$ as the glint panel unit normal in the inertial frame. The ideal specular reflection direction is given by $\mathbf{r}_{\text{spec}} = 2(\mathbf{n}_g \cdot \hat{\mathbf{s}}_{\odot})\mathbf{n}_g - \hat{\mathbf{s}}_{\odot}$.

The angle ψ between this ideal specular direction and the observer line-of-sight $\hat{\mathbf{e}}$ is given by $\cos \psi = \mathbf{r}_{\text{spec}} \cdot \hat{\mathbf{e}}$. In (7) ψ acts as a misalignment angle, controlling the attenuation of glint intensity with observer offset from the specular direction and σ parameterize the surface effective specular beam-width i.e. small values correspond to a highly polished, narrow-beam reflector, while large values model broader scattering from imperfectly flat or rough surfaces.

Eqs. (5)–(7) fully define the model flux given the geometry, rotation phase, and viewing conditions. All surface normal vectors, including those of the bus faces and the glinting panel, are initially defined in the satellite body frame. For flux computations, these normals are transformed into the inertial (ICRF) frame at the corresponding rotation phase. The transformation consists of a fixed alignment of the body-frame z -axis with the spin axis $\alpha_{\text{pole}}, \delta_{\text{pole}}$, followed by a rotation about the spin axis according to the instantaneous rotation angle. In the following, we will assume that the rotation axis is fixed in inertial space, i.e., $\alpha_{\text{pole}}, \delta_{\text{pole}}$ remain constant throughout the entire fitting period.

The model fitting procedure operates on mean-corrected lightcurves stored in plain-text input files. Each file contains the topocentric right ascension and declination of the satellite, the observation epoch and the photometric time-series data, preprocessed to remove any constant flux offset. This ensures that the fitting focuses on lightcurve amplitude modulated by the satellite rotation rather than absolute brightness calibration. For computational efficiency, each lightcurve is binned into 100 equally spaced bins before fitting. The bin size is well below the lightcurve variability timescale, thus reducing the computational cost of the model fitting while preserving the lightcurve morphology. This normalization and binning ensure that the diffuse and glint flux parameters A_{box} and G are fitted purely from relative variations in the rotational modulation, making them robust to unknown absolute albedo and cross-section scaling.

The determination of the model parameters from the lightcurves constitutes a high dimensional, nonlinear optimization problem. The model includes geometric parameters (spin-pole coordinates, bus axis ratios, glinting panel orientation) and photometric scaling parameters (diffuse and specular amplitudes, glint width), many of which may exhibit strong correlations. The dynamical landscape of the problem is complicated by the fact that different combinations of shape, orientation, and glint parameters can produce similar lightcurve shapes, and by the existence of symmetries in the bus geometry (e.g., 90° rotations) and the periodicity of the rotational phase, both of which generate multiple equivalent local minima [58].

In the specific case of our satellite lightcurve model, these geometric symmetries and the interplay between diffuse and specular components lead to geometry-driven degeneracies: multiple distinct parameter sets that yield nearly identical lightcurve signatures. This makes global exploration essential to avoid settling on a non-physical local minimum. To address this, we employ a two-step approach. First, we perform a global optimization using the Differential Evolution (DE) algorithm [62], a population-based, stochastic search method particularly effective for non-convex optimization problems. DE explores a wide parameter space by iteratively mutating and recombining parameter vectors, retaining the best-performing candidates according to the cost function. In the second stage, the best DE solution is used to initialize the Levenberg–Marquardt (LM) algorithm [63] for local refinement. The LM stage, a non-linear least-squares solver, rapidly converges to the nearest local minimum, providing the final fitted parameters of the satellite lightcurve model. Both DE and LM were used via their respective implementations in the SciPy library [64], ensuring robust and well-tested numerical performance. This hybrid DE+LM strategy ensures that the global topology of the problem is explored first, mitigating the risk of convergence to a non-physical local minimum, while still benefiting from the precision and efficiency of a local optimization algorithm in the final stage.

For both optimization stages, the optimization is driven by a reduced chi-squared cost function of parameter set $\mathbf{p}$, evaluated over the $N = 100$ points of the binned, epoch-folded lightcurve:

$$\chi_v^2(\mathbf{p}) = \frac{1}{N_{\text{DOF}}} \sum_{i=1}^N \left[\frac{M_{m,i} - M_{s,i}(\mathbf{p})}{\sigma_i} \right]^2, \quad (8)$$

where $M_{m,i}$ and $M_{s,i}(\mathbf{p})$ are the observed and model-predicted magnitudes for bin i , σ_i is the corresponding observational uncertainty, and $N_{\text{DOF}} = N - N_{\text{par}}$ is the number of degrees of freedom, with N_{par} the total number of fitted parameters. All model computations are carried out internally in *relative flux units*, ensuring that the diffuse and specular components combine linearly. The conversion to magnitudes is performed only for the evaluation of the cost function.

The method simultaneously fits multiple lightcurves obtained at different epochs, sharing a single set of global parameters (spin pole, bus axis ratios, panel orientation, $A_{\text{box}}, G, \sigma$) while assigning a separate phase offset to each lightcurve. Although the rotation period P is measured with a relative precision of 10^{-3} , the cumulative phase drift over the many rotations separating the epochs makes the rotational phase at the start of each lightcurve effectively unknown.

We therefore model the instantaneous rotational phase of sample (magnitude measurement) j in lightcurve k as

$$\theta_{k,j} = 2\pi \varphi_{k,j} + \Delta\phi_k,$$

where $\varphi_{k,j} \in [0, 1)$ is the fractional phase within that lightcurve (after binning and epoch folding), and $\Delta\phi_k \in [0, 2\pi)$ is a free, per-lightcurve phase offset. The term $\Delta\phi_k$ absorbs the unknown absolute orientation at that epoch as well as any accumulated drift from small period errors. This formulation allows all epochs to share a single set of physical and geometric parameters (e.g., spin-axis orientation, panel normals, and shape ratios) while remaining insensitive to phase drifts. By combining lightcurves observed under different Sun-observer geometries, the method also reduces parameter degeneracies, since each geometry samples the scattering behavior of the satellite from a different aspect.

The DE optimization was run with a population size of 120 individuals for 600 generation. This configuration was chosen to provide a thorough exploration of the multidimensional parameter space while keeping a reasonable computational cost (45 min on an Apple M1 Max, 10-core CPU).

Following the DE stage, the entire final population was stored in order to examine the local density of solutions in parameter space. The presence of a single, compact cluster in parameter space is indicative of good convergence and suggests that the solution is well constrained within the model assumptions. Conversely, the existence of multiple high density regions indicates the degeneracies in the model, where different parameter combinations can reproduce the observed lightcurves equally well. In addition, the final DE population provides a direct means of estimating parameter uncertainties and correlations.

Our observational dataset, presented in Appendix A, comprises nine satellites, each with more than ten individual lightcurves. Although a comprehensive analysis of the full sample is beyond the scope of this work and will be presented in a subsequent paper, we focus here on a detailed case study to demonstrate the methodology. To this end, we selected Raduga 3 (1977-071A), one of our targets with extended observational coverage, making it an ideal benchmark for testing our approach. Raduga 3 was a military communications satellite and the first to employ the KAUR-3 spacecraft bus.³ The KAUR-3 platform featured three-axis stabilization using liquid-propellant micro-engines, achieving pointing accuracy within 0.25° of Earth's center, and incorporated a gyroscope-based attitude control system in combination with micro-thrusters. Its solar arrays could independently track the Sun, eliminating the need to reorient the spacecraft in orbit. Raduga 3 had a cylindrical body with two solar panels, measuring 5.5 m in height, and 2.5 m in diameter.

Fig. 16 shows the individual fits to each of the four lightcurves of 1977-071A. It is evident that the simple model, consisting of a diffuse bus and a single specular panel, can reproduce the asymmetric features of the observed lightcurves. By fitting together the two lightcurves from June and the two from July, we still obtain a reasonably good fit. However, the residuals begin to show a periodic structure, indicative of incomplete modeling of the actual satellite geometry (Fig. 17). This points to the presence of multiple reflective surfaces contributing to the observed lightcurves. The simultaneous fit across four lightcurves obtained over one month (Fig. 18) captures the broad characteristics of the brightness variation but presents clear, periodic residuals. This is expected given the relatively simple modeling of the satellite with the bus represented by six Lommel–Seeliger faces with azimuth-independent reflectance, self-shadowing and mutual occlusions ignored, and the specular behavior approximated for a single panel with a Gaussian angular response. Real spacecraft often present multiple reflective elements (arrays, radiators, antennae) and misaligned panels. Any of these effects can introduce asymmetric peaks and narrow features that the current model cannot fully render. Consequently, the fit should be interpreted as capturing the dominant, physically relevant trends of the lightcurves with systematic residuals that reflect unmodeled geometry.

Finally, we assess the quality and stability of the global fit by analyzing the clustering of the DE population in the final parameter space. A compact, well-defined cluster indicates good convergence and suggests that the solution is robust, while multiple dense regions or elongated structures in parameter space may reflect existing degeneracies in the lightcurve model. Fig. 19 presents the final distribution of DE population in the $(\alpha_{\text{pole}}, \delta_{\text{pole}})$ parameter space, with density contours showing the 50 %, 80 % and 95 % confidence regions. The solutions are clustered around $\alpha_{\text{pole}} \approx 330^\circ$, $\delta_{\text{pole}} \approx 75^\circ$. The LM refinement step found $\alpha_{\text{pole}} = 331.7 \pm 2.4^\circ$, $\delta_{\text{pole}} = 74.8 \pm 0.5^\circ$. We note that the 80 % confidence region solution (Fig. 19) partially overlaps with our previous finding on 1977-071 pole location (Fig. 15).

Another important result concerning the satellite geometry comes from the distribution of the glinting panel co-latitude θ_g and azimuth ϕ_g in the DE population (Fig. 20). The pronounced clustering at low co-latitude values indicates a panel oriented nearly perpendicular to the rotation axis (Fig. 21 a). Finally, the relative dimensions of the bus are inferred from the b/a and c/a ratios obtained from the DE population, providing constraints on the satellite overall shape. In our model, the satellite rotation axis is aligned with the z -axis, meaning that the faces perpendicular to the rotation axis are the $\pm Z$ faces. These faces have sides a and b , based on the normalized shape ratios $a : b : c = 3.0 : 1.0 : 7.2$. The c dimension, being the largest, lies parallel to the rotation axis, defining the bus's length, while the $\pm Z$ faces form its ends. This geometry is consistent with the interpretation that the glinting panel, located at a low co-latitude are mounted nearly perpendicular to the spin axis. These findings are compatible with known specifications of the KAUR-3 platform used for Raduga spacecraft.

Extending the global fit model to four glinting panels led to a decrease of lightcurves residuals (Fig. 18) while the overall rms error was reduced from 0.359 to 0.289 (Appendix B). The normalized dimensions of the bus are $a : b : c = 2.0 : 1.0 : 5.2$, which are closer to the KAUR-3 specifications than those derived from the single-panel case. In this configuration, three panels with similar photometric properties (weak specular strength) are located at low co-latitudes ($\approx 20^\circ$), consistent with the results obtained from the single-panel model. However, a fourth panel with a strong glint component was identified at a co-latitude of $\approx 160^\circ$, i.e., oriented toward the $-Z$ direction (Fig. 21 b). This demonstrates that multiple reflective elements are necessary to capture the observed photometric complexity, with the single panel model providing only a first order characterization and the four

³ <https://www.russianspaceweb.com/raduga.html>

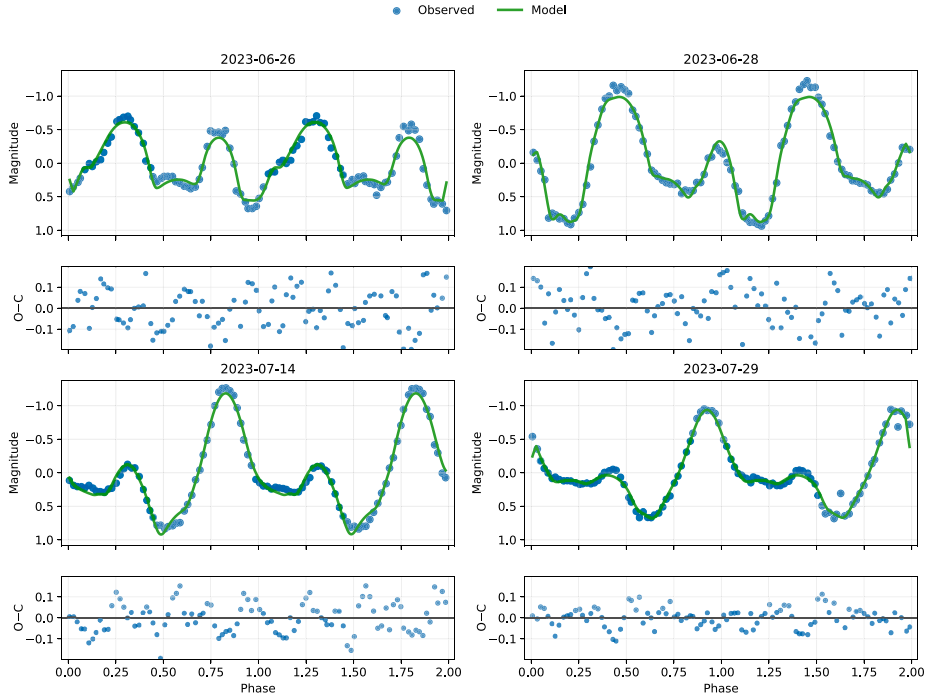


Fig. 16. Individual fit of 4 lightcurves of 1977-071A.

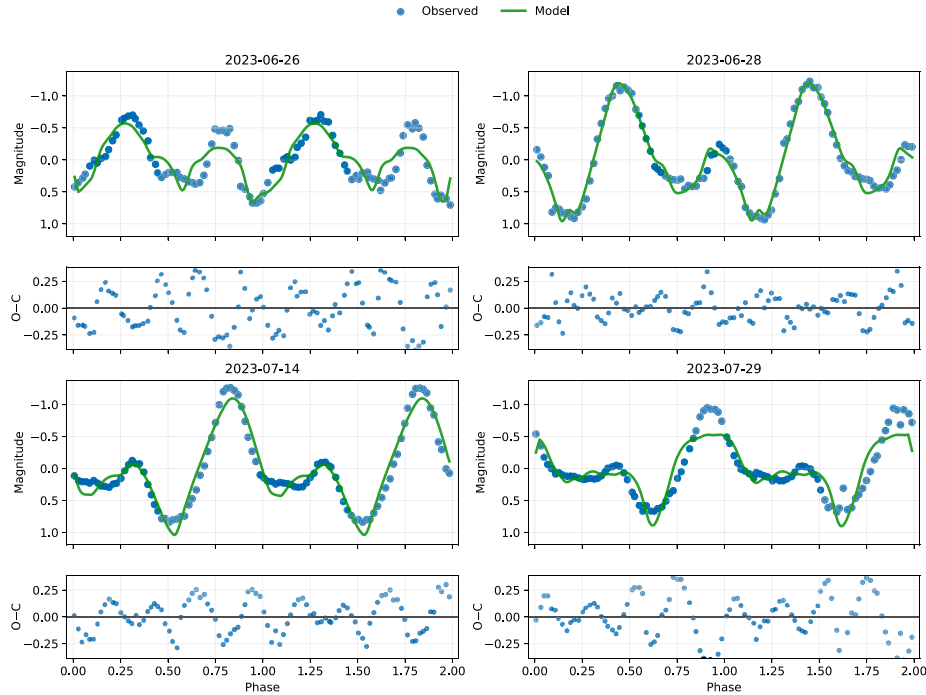


Fig. 17. Global fit of June (upper panel) and July (lower panel) lightcurves.

panel model yielding a higher fidelity reconstruction of the rotational lightcurves. We do not exclude the possibility that part of the remaining residual structure arises from the simplified assumption of a fixed rotation axis. Slow precession or nutation of the spin axis, induced by asymmetric solar radiation pressure torques or internal mass distribution, could produce additional variability not captured by the present model. We note, however, that both approaches converge on the same orientation of the rotation axis and similar overall dimensions of the bus. In the four panel configuration, all reflective surfaces are oriented at relatively low co-latitude, i.e. nearly perpendicular to the

spin axis. Three panels are oriented toward the $+Z$ direction, while the strong glinting component faces the $-Z$ direction. This configuration may point to extended structures such as solar arrays or radiators aligned along the body's long dimension. Their azimuths display a nearly symmetric arrangement about the spin axis, further supporting the interpretation of repeated or paired structural elements such as two solar panels or radiator wings mounted on opposite sides of the spacecraft. The four panel model has an increased computational time, since it introduces twelve additional free parameters compared to the single-panel case.

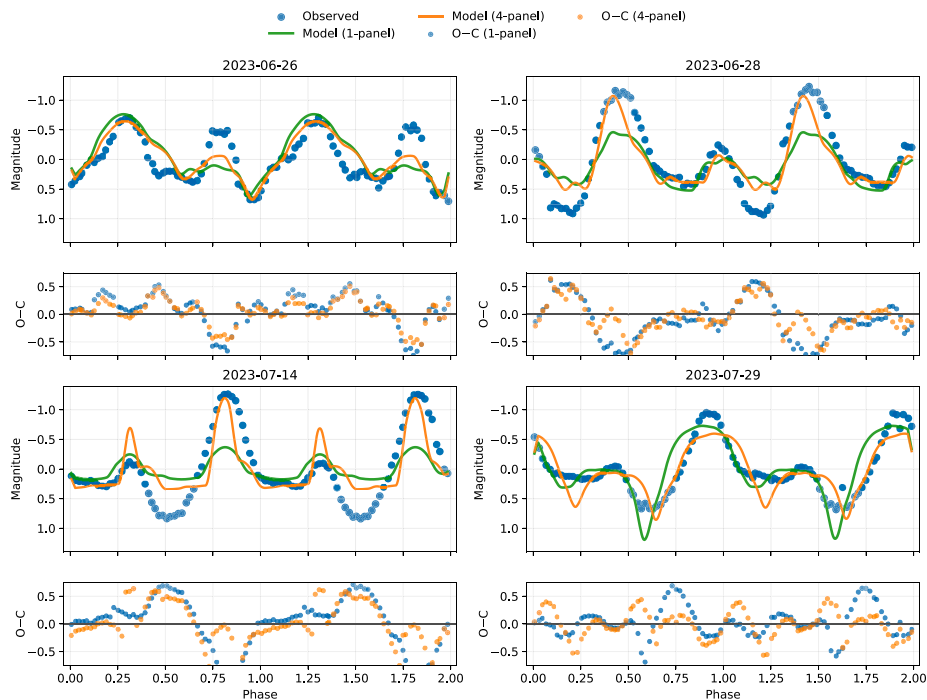


Fig. 18. Global fit of all 4 lightcurves using one panel (green) and four panels (orange) models. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

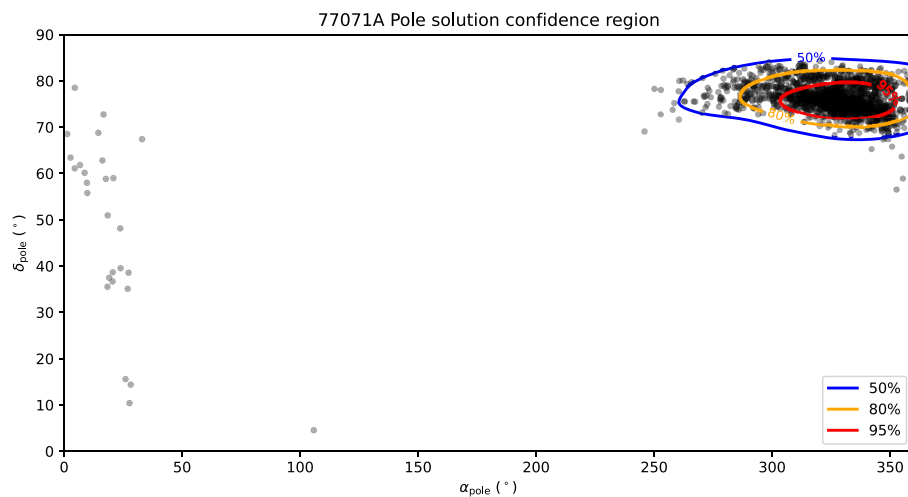


Fig. 19. Confidence regions for the spin-pole solution of 1977-071A derived from the clustering of DE final population in $(\alpha_{\text{pole}}, \delta_{\text{pole}})$ space.

4. Conclusions

This paper presents the APSIS program, a high precision long term photometric monitoring of GEO and MEO objects, producing a unique multi-year photometric dataset. This capability enables the systematic study of satellite spin dynamics and provides model ready parameters for space situational awareness. At the core of APSIS is a fully automated pipeline that handles the complete chain from image acquisition to science-ready products. The system autonomously schedules observations, detects and tracks objects, calibrates raw frames, performs astrometric and photometric reduction, producing calibrated lightcurves and high precision astrometry in near real time. Rotation periods are automatically computed and archived, ensuring that the

dataset is continuously updated without the need for manual intervention. This high level of automation enables APSIS to maintain consistent quality across large volumes of data, making it possible to monitor a wide range of targets over multi-year timescales. For a subset of nine satellites continuously monitored with APSIS, we identified periodic variations in their rotation periods with a near-annual timescale. We interpret these modulations as being driven by the annual variation of solar radiation pressure torques, whose magnitude depends on the geometry between the Sun and the satellite's spin axis. By fitting the observed period evolution with a torque-driven model, we obtained families of admissible spin-axis orientations. Although the intrinsic symmetries of the problem prevent unique pole solutions, the analysis demonstrates quantitative consistency between SRP forcing and the

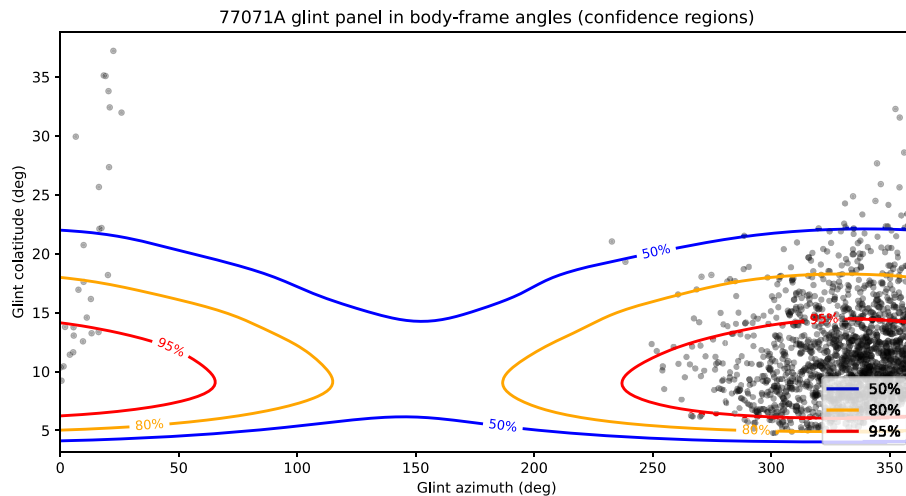


Fig. 20. Confidence regions for the specular panel of 1977-071A orientation in body frame derived from the clustering of DE final population in θ_g, ϕ_g space.

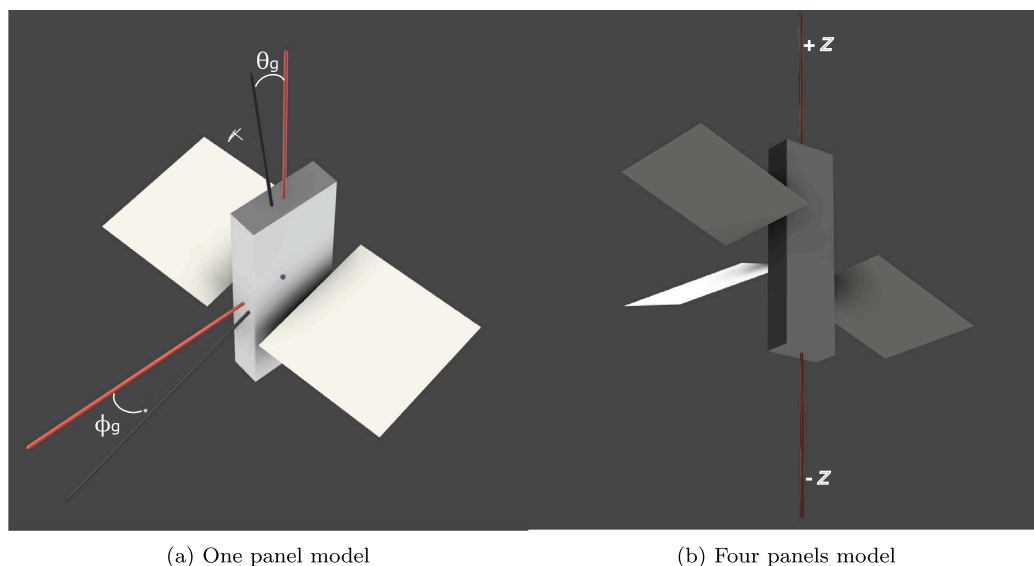


Fig. 21. The derived generic shape of 1977-071A from photometric lightcurve inversion. One panel (a) with co-latitude $\theta_g = 7^\circ$ and azimuth $\phi_g = 351^\circ$. Four panels (b) with three low co-latitude panels represented as a single one and a fourth strong glinting component facing $-Z$ direction, i.e. $\theta_g = 158^\circ$. Only bus dimensions are to scale.

measured spin dynamics, and constrains the spin axis to well-defined regions on the sky.

Furthermore, we developed and applied a physically motivated lightcurve inversion framework for characterizing satellite geometry from unresolved optical photometry. The model describes the spacecraft as a diffuse-reflecting rectangular bus with six faces following the Lommel–Seeliger law, complemented by a single specular panel with a Gaussian bidirectional reflectance profile. The geometry is defined by a minimal set of physically interpretable parameters: spin-pole orientation, bus axis ratios, glint panel orientation, and photometric scaling terms. Using a two-stage optimization procedure, Differential Evolution (DE) for global exploration, followed by Levenberg–Marquardt (LM) refinement, we jointly fit multiple phase-binned, zero-mean lightcurves obtained at different epochs. The DE final population provides insight into parameter uncertainties and reveals the structure of the solution space, highlighting both well-constrained parameters and existing degeneracies driven by geometric symmetries.

Application to a set of four lightcurves of the geostationary satellite 1977-071A (Raduga/KAUR-3 class) demonstrates that the model reproduces the main asymmetric features of the observed lightcurves. The best-fit parameters indicate a glinting panel nearly perpendicular to the spin axis and bus axis ratios consistent with an elongated body $a : b : c = 3.0 : 1.0 : 7.2$. These values are broadly compatible with known KAUR-3 platform dimensions (5.5 m length, 2.5 m diameter). Extending the model to four glinting panels further reduced residuals and recovered a nearly symmetric arrangement of reflective surfaces at low co-latitudes, consistent with paired or repeated structural elements such as solar arrays or radiators, while converging to the same pole solution and overall dimensions of the bus.

Despite its simplicity, the model successfully links lightcurves obtained weeks apart, with systematic residuals reflecting the absence of effects such as self-shadowing, mutual facet occlusion, and multiple specular elements. Incorporating these processes in future extensions could improve fit accuracy and reduce systematic residuals, further enhancing the diagnostic power of unresolved satellite photometry.

CRediT authorship contribution statement

Mădălina Trelia: Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dan Alin Nedelcu:** Writing – original draft, Validation, Software, Methodology. **Mirel Birlan:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Observation circumstances

See [Table A.2](#).

Appendix B. Model parameters

See [Table B.3](#).

Table A.2

Columns are in order: observations interval, phase angle range, period (s) and the associated error (ms) determined using the Lomb-Scargle method and the period determined using the epoch folding method and the lightcurve amplitude.

1974-060A						
Observation period		Φ range	P	σ_p	P^1	Amp
Date	Time	(°)	(s)	(ms)	(s)	(mag)
2023-06-23	22:10:35 - 23:03:33	33.3 - 44.9	25.699	35	25.698	1
2023-06-26	21:15:12 - 21:44:44	23.0 - 28.5	25.699	66	25.695	0.8
2023-06-28	23:04:56 - 23:34:49	46.0 - 52.8	25.698	64	25.685	1
2023-06-29	21:03:51 - 21:33:36	21.5 - 26.7	25.701	64	25.695	0.8
2023-07-08	21:39:32 - 22:10:53	29.2 - 35.7	25.704	62	25.705	0.8
2023-07-11	19:53:02 - 20:40:21	16.1 - 19.2	25.705	43	25.708	0.7
2023-07-14	19:28:03 - 20:33:02	17.1 - 18.4	25.706	33	25.706	0.6
2023-07-15	19:41:46 - 20:27:11	16.0 - 17.8	25.705	45	25.709	0.7
2023-07-16	20:34:17 - 21:06:02	18.6 - 23.7	25.712	61	25.710	0.7
2023-07-18	19:36:35 - 20:23:35	15.9 - 17.4	25.710	42	25.709	0.6
2023-07-24	21:03:18 - 22:08:38	23.9 - 37.6	25.714	29	25.712	0.8
2023-07-29	21:59:42 - 22:14:45	36.4 - 39.8	25.711	133	21.713	0.9
2023-08-17	21:06:27 - 21:22:36	27.8 - 31.5	25.709	63	25.709	1.1
2023-08-21	20:36:34 - 21:10:13	21.7 - 29.5	25.707	29	25.707	1.6
2023-09-11	21:01:02 - 21:08:47	33.4 - 35.3	25.706	248	25.711	0.7
2023-10-07	19:08:25 - 19:37:37	16.8 - 23.5	25.709	68	25.705	0.8
2025-01-16	16:29:33 - 16:45:25	60.7 - 57.3	25.655	110	25.653	1
2025-01-26	17:40:47 - 17:56:24	47.6 - 44.5	25.661	90	25.656	1.6
2025-02-10	17:35:07 - 17:52:53	50.2 - 46.4	25.666	91	25.665	1.9
2025-02-21	22:42:19 - 22:59:57	35.1 - 38.8	25.663	113	25.665	0.8
2025-03-02	23:06:01 - 23:23:39	37.6 - 41.6	25.666	121	25.668	2.5
2025-03-05	18:55:28 - 19:12:58	32.4 - 28.7	25.665	95	25.667	1.6
2025-05-26	19:36:49 - 19:54:22	24.3 - 20.9	25.688	121	25.690	0.5
2025-06-02	20:49:20 - 21:06:53	15.1 - 14.9	25.691	118	25.686	0.6
2025-06-08	20:23:59 - 20:33:18	18.1 - 17.1	25.678	221	25.684	0.6
2025-06-21	20:47:50 - 21:09:44	17.2 - 16.9	25.695	100	25.698	0.7
2025-07-05	21:02:02 - 21:36:57	16.7 - 18.7	25.699	61	25.698	0.7
1977-071A						
Observation period		Φ range	P	σ_p	P^1	Amp
Date	Time	(°)	(s)	(ms)	(s)	(mag)
2023-03-13	20:56:18 - 21:27:22	17.7 - 21.9	12.221	14	12.222	1.1
2023-03-18	19:06:57 - 19:30:51	25.5 - 21.0	12.185	18	12.187	1.2
2023-03-19	17:34:56 - 18:00:26	46.2 - 40.2	12.179	17	12.178	2.7
2023-03-21	23:41:55 - 23:53:41	47.9 - 50.7	12.151	45	12.166	1.1
2023-03-24	21:18:49 - 21:45:29	16.1 - 20.9	12.135	15	12.136	1
2023-06-04	20:10:56 - 20:24:41	19.1 - 16.5	11.666	33	11.663	1.1
2023-06-10	21:03:09 - 21:17:35	12.7 - 12.2	11.648	28	11.650	2.5
2023-06-20	20:19:36 - 20:52:12	20.6 - 15.7	11.640	12	11.641	4.6
2023-06-26	22:22:33 - 22:50:58	20.0 - 25.2	11.648	14	11.649	1.4
2023-06-28	23:41:40 - 00:14:31	35.8 - 43.1	11.656	12	11.655	2.2
2023-07-15	22:33:05 - 23:11:51	21.1 - 28.5	11.735	10	11.734	2.2
2023-07-29	22:22:51 - 22:37:54	18.3 - 26.5	11.845	56	11.843	1.6
2023-09-05	19:28:50 - 20:07:24	30.1 - 20.8	12.197	23	12.197	1.5
2023-10-06	19:31:32 - 19:59:09	24.1 - 17.4	12.490	34	12.491	1.4
2023-10-07	21:59:32 - 22:26:08	14.8 - 21.2	12.495	35	12.488	1.1
2023-10-23	19:39:27 - 20:19:03	21.0 - 13.6	12.659	24	12.658	1.3
2023-11-16	17:23:33 - 17:42:20	50.1 - 45.8	12.917	54	12.917	1.4

(continued on next page)

Table A.2 (continued).

2023–11–23	20:10:55 - 20:29:43	21.8 - 21.8	12.988	53	12.987	1.3
2023–12–16	17:22:08 - 17:40:56	48.0 - 44.2	13.111	54	13.110	1.6
2024–01–02	20:43:32 - 21:02:14	31.5 - 33.6	13.068	26	13.066	1.7
2024–01–05	19:16:20 - 19:55:02	29.8 - 28.7	13.046	27	13.047	2.1
2024–01–10	18:58:19 - 19:40:58	30.9 - 28.7	13.010	23	13.012	1.9
2024–01–21	18:47:17 - 19:05:53	30.5 - 29.0	12.922	25	12.917	1.5
2025–06–02	19:51:48 - 20:09:20	16.1 - 18.9	12.285	17	12.283	3.2
2025–06–08	19:42:36 - 20:00:13	14.9 - 17.0	12.277	15	12.279	3
2025–06–21	20:01:57 - 20:24:07	16.7 - 19.2	12.311	20	12.311	1.7
2025–07–01	20:13:33 - 20:48:23	17.0 - 21.3	12.373	19	12.373	2.1
2025–07–05	23:26:14 - 00:01:09	53.1 - 61.1	12.405	14	12.405	1.7
1979-015A						
Observation period		Φ range	P	σ_P	P^1	Amp
Date	Time	($^{\circ}$)	(s)	(ms)	(s)	(mag)
2023–07–27	19:26:52 - 20:55:32	19.6 - 15.6	84.610	245	84.620	2.1
2023–07–29	22:58:59 - 23:29:49	40.2 - 47.4	84.519	694	84.553	1.3
2023–09–05	20:09:24 - 20:48:06	6.90 - 8.80	81.901	266	81.924	0.6
2023–09–11	20:15:26 - 20:56:48	4.50 - 10.1	81.600	240	81.628	0.6
2023–10–13	18:08:15 - 18:27:03	29.8 - 25.1	80.709	524	80.729	1.9
2023–10–23	21:12:25 - 21:50:34	21.7 - 30.5	80.363	248	80.387	0.8
2023–11–02	16:31:05 - 17:09:17	52.0 - 42.7	80.180	256	80.178	1.5
2023–11–02	19:03:56 - 19:34:18	17.7 - 14.0	80.152	316	80.155	0.7
2023–12–16	17:43:12 - 18:02:02	35.9 - 32.8	78.893	496	78.939	1.4
2023–12–29	16:27:03 - 16:45:50	50.6 - 46.7	78.467	459	78.449	1.4
2024–01–14	18:30:20 - 19:10:34	29.7 - 27.6	77.333	452	77.360	1.3
2024–01–21	16:55:22 - 17:34:08	43.2 - 36.3	76.699	451	76.720	1.2
2025–06–08	20:02:59 - 20:19:17	14.6 - 16.9	67.218	360	67.215	1.1
2025–06–21	20:24:42 - 20:46:36	17.7 - 21.2	66.465	207	66.460	1.2
2025–06–22	20:01:32 - 20:23:26	15.2 - 17.5	66.394	298	66.404	1.1
2025–06–29	21:22:17 - 21:56:02	27.2 - 34.4	66.008	192	66.024	1.4
2025–07–05	22:50:41 - 23:25:41	45.7 - 53.8	65.704	196	65.694	1.4
1983-100A						
Observation period		Φ range	P	σ_P	P^1	Amp
Date	Time	($^{\circ}$)	(s)	(ms)	(s)	(mag)
2023–05–22	18:53:03 - 19:37:32	18.9 - 8.50	12.981	13	12.981	1.1
2023–06–26	23:07:59 - 23:37:42	49.5 - 56.6	12.981	18	12.981	4.6
2023–06–29	00:16:43 - 00:31:14	66.5 - 70.0	12.981	68	12.979	2.4
2023–06–29	21:38:28 - 22:10:37	28.9 - 36.4	12.979	15	12.980	1.6
2023–07–11	20:44:04 - 21:15:25	19.1 - 25.9	12.970	17	12.973	4.2
2023–07–14	20:49:05 - 21:38:42	20.7 - 31.7	12.966	10	12.965	3.7
2023–07–15	20:37:20 - 21:50:14	18.4 - 34.6	12.965	6	11.343	4.6
2023–10–07	20:10:45 - 21:07:13	30.0 - 44.3	12.879	18	12.878	1.6
2025–01–14	16:42:23 - 16:58:18	52.3 - 48.8	12.580	25	12.580	1.8
2025–01–15	16:36:44 - 16:52:31	53.8 - 50.3	12.581	28	12.580	1.9
2025–01–25	20:03:39 - 20:19:33	24.6 - 25.1	12.586	34	12.587	1.8
2025–02–04	18:31:24 - 18:49:02	33.4 - 30.5	12.588	28	12.589	4.9
2025–02–11	17:05:23 - 17:22:55	51.7 - 47.8	12.588	28	12.588	5.6
2025–02–21	22:18:59 - 22:36:30	35.8 - 39.3	12.586	9	12.587	1.4
2025–03–02	23:27:21 - 23:44:58	48.3 - 52.3	12.589	12	12.590	1.6
2025–05–26	19:55:59 - 20:13:21	10.3 - 7.3	12.569	25	12.570	3.6
2025–06–21	19:37:02 - 19:58:59	15.8 - 12.4	12.567	16	12.568	1.8
2025–06–29	20:43:26 - 21:18:11	12.5 - 17.5	12.562	15	12.562	1.6
2025–07–11	20:14:23 - 20:58:48	12.2 - 15.3	12.551	13	12.551	4
1985-102A						
Observation period		Φ range	P	σ_P	P^1	Amp
Date	Time	($^{\circ}$)	(s)	(ms)	(s)	(mag)
2023–12–27	15:59:47 - 16:18:37	54.1 - 49.7	18.912	56	18.916	1
2024–01–02	21:05:43 - 21:24:36	34.4 - 38.3	18.917	58	18.920	1.3
2024–01–11	16:32:43 - 16:51:55	45.4 - 41.2	18.915	111	18.918	1.1
2024–01–14	17:49:59 - 18:28:29	29.2 - 23.5	18.916	14	18.918	1.1
2024–01–20	17:13:47 - 17:52:01	35.7 - 28.4	18.906	14	18.906	1.2
2024–01–30	21:13:40 - 21:37:40	38.1 - 43.2	18.879	47	18.884	1.5
2025–01–17	17:30:23 - 17:46:17	21.8 - 21.5	18.853	62	18.851	1.3
2025–01–27	16:55:59 - 17:13:29	25.3 - 23.1	18.853	52	18.836	1.5
2025–02–25	17:16:24 - 17:33:56	22.5 - 20.0	18.812	53	18.808	1.9
2025–06–22	19:37:57 - 19:59:53	8.3 - 11.3	18.785	71	18.785	1.6
2025–06–30	20:12:01 - 20:25:41	13.1 - 15.7	18.787	109	18.784	1.5
2025–06–30	20:31:54 - 21:06:51	17.0 - 24.7	18.783	40	18.784	1.6
2025–07–05	22:15:03 - 22:50:06	40.1 - 48.5	18.789	42	18.788	2.6

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Table A.2 (continued).

1990-051A						
Observation period		Φ range	P	σ_P	P^1	Amp
Date	Time	(°)	(s)	(ms)	(s)	(mag)
2023-03-13	17:39:58 - 17:56:15	22.4 - 18.4	10.052	9	10.052	2.7
2023-03-24	18:33:47 - 18:45:55	8.90 - 6.3	9.950	12	9.948	3.1
2023-06-29	23:59:43 - 00:48:17	71.4 - 83.3	9.267	10	9.266	3.2
2023-07-29	23:32:41 - 23:47:40	66.7 - 70.5	9.315	9	9.316	3.8
2023-09-11	21:01:02 - 21:43:10	33.3 - 43.8	9.540	3	9.539	2.8
2023-10-06	20:01:54 - 21:09:45	22.1 - 38.4	9.694	2	9.694	2.9
2023-10-14	21:48:56 - 22:08:02	49.0 - 53.8	9.745	8	9.745	4
2023-12-16	18:02:57 - 18:21:48	15.8 - 14.6	10.017	8	10.017	2.2
2023-12-27	16:20:24 - 16:39:02	35.2 - 30.9	9.990	7	9.989	2.1
2024-01-10	20:02:14 - 20:21:00	28.5 - 32.8	9.910	8	9.909	5.2
2024-01-11	16:54:15 - 17:12:59	26.5 - 22.5	9.903	8	9.904	2.2
2024-01-14	20:16:41 - 20:55:25	31.8 - 41.0	9.880	4	9.880	2.4
2024-01-20	21:51:16 - 22:29:46	54.6 - 64.0	9.832	4	9.832	3.4
2024-01-22	16:50:40 - 18:27:14	26.5 - 12.4	9.817	2	9.818	1.8
2024-01-22	19:06:13 - 20:49:12	16.7 - 39.4	9.817	1	9.817	3.4
2024-11-01	19:27:02 - 19:35:45	18.0 - 19.7	9.256	21	9.256	5.5
2024-11-01	19:40:09 - 19:52:36	20.6 - 23.2	9.254	16	9.259	3.3
2024-11-15	22:40:53 - 23:20:30	63.6 - 73.5	9.316	11	9.317	2.5
2024-11-16	17:36:07 - 18:16:31	20.8 - 14.5	9.320	9	9.319	2.7
2024-11-17	17:36:52 - 18:17:19	20.7 - 14.5	9.324	7	9.323	1.7
2025-01-02	18:57:56 - 19:25:19	14.0 - 16.5	9.303	5	9.302	1.7
2025-01-03	21:16:08 - 21:32:03	39.6 - 43.4	9.297	8	9.297	5.8
2025-01-08	17:59:06 - 18:09:38	19.2 - 17.4	9.270	13	9.270	2.3
2025-01-09	17:19:05 - 17:48:09	27.5 - 21.3	9.264	4	9.264	3.1
2025-01-11	18:28:01 - 18:43:55	14.8 - 13.6	9.252	8	9.252	2.2
2025-01-14	17:01:51 - 17:17:44	31.5 - 27.9	9.233	8	9.233	3.1
2025-01-26	20:56:59 - 21:31:08	33.1 - 41.2	9.149	4	9.149	3.1
2025-02-10	21:38:31 - 21:53:45	41.7 - 45.4	9.042	9	9.041	3.6
2025-02-19	18:02:55 - 18:20:24	17.5 - 14.0	8.977	7	8.977	5.5
2025-03-05	17:40:14 - 17:57:52	22.2 - 18.0	8.878	7	8.879	2.8
1994-008A						
Observation period		Φ range	P	σ_P	P^1	Amp
Date	Time	(°)	(s)	(ms)	(s)	(mag)
2023-03-26	17:46:15 - 18:13:31	28.9 - 22.2	8.871	8	8.871	1.5
2023-05-22	18:52:24 - 19:47:28	19.9 - 6.6	8.822	4	8.822	2.4
2023-05-23	19:30:14 - 19:48:47	10.9 - 6.4	8.825	13	8.823	2.5
2023-05-27	00:19:48 - 00:34:18	58.2 - 61.7	8.821	17	8.820	3
2023-06-20	22:26:01 - 22:57:30	28.5 - 36.0	8.819	7	8.819	2.7
2023-06-26	21:49:14 - 22:18:53	19.2 - 26.3	8.820	16	8.820	2.4
2023-07-29	22:41:39 - 22:56:42	30.8 - 34.5	8.823	31	8.821	3.6
2023-08-21	19:48:06 - 20:04:14	11.3 - 7.5	8.818	28	8.819	2.8
2023-09-06	21:59:47 - 22:38:38	22.8 - 32.5	8.810	6	8.810	1.8
2023-09-11	19:30:15 - 20:00:58	15.1 - 8.3	8.810	7	8.809	1.7
2023-10-07	18:28:37 - 19:06:52	28.5 - 19.6	8.785	6	8.785	1.6
2023-11-01	18:19:01 - 18:38:11	28.9 - 24.5	8.744	12	8.744	1.3
2023-11-02	18:55:58 - 19:34:18	20.5 - 13.5	8.742	6	8.742	1.2
2023-12-27	15:59:47 - 16:18:37	54.8 - 50.2	8.608	12	8.608	3
2024-01-02	20:20:36 - 20:39:49	20.7 - 24.5	8.591	11	8.591	1.4
2024-01-10	19:42:01 - 20:01:13	15.4 - 18.1	8.569	11	8.569	1.2
2024-01-14	19:56:34 - 20:15:21	17.7 - 21.1	8.558	12	8.558	1.3
2024-01-20	18:02:29 - 18:21:00	21.3 - 17.9	8.543	12	8.543	1.3
2024-02-08	21:20:42 - 22:01:07	39.3 - 49.1	8.493	5	8.493	2.8
2024-03-03	17:34:28 - 18:14:00	16.7 - 9.7	8.443	5	8.843	1.2
2024-11-16	18:17:22 - 18:57:47	23.4 - 32.5	8.269	5	8.269	2.6
2025-01-09	16:44:46 - 17:17:06	19.4 - 15.2	8.149	5	8.148	1.8
2025-01-11	18:07:50 - 18:23:48	16.5 - 18.8	8.143	11	8.143	1.8
2025-01-26	17:23:35 - 17:38:55	14.9 - 13.6	8.110	11	8.109	3.4
2025-03-05	17:59:34 - 18:17:58	10.6 - 9.20	8.037	8	8.037	1.3
2025-07-05	19:52:29 - 20:27:43	2.70 - 6.6	7.982	6	7.982	3.4
1994-087A						
Observation period		Φ range	P	σ_P	P^1	Amp
Date	Time	(°)	(s)	(ms)	(s)	(mag)
2023-12-16	18:03:02 - 18:21:48	15.5 - 14.5	45.670	343	45.702	1.1
2023-12-27	16:20:07 - 16:39:02	34.6 - 30.3	46.515	350	46.534	1.5
2024-01-10	20:02:14 - 20:20:48	29.0 - 33.2	47.695	364	47.686	1.6
2024-01-11	16:54:15 - 17:12:53	26.3 - 22.4	47.768	381	47.764	1.3
2024-01-14	19:11:39 - 19:30:47	18.5 - 22.1	47.949	368	48.008	1.3
2024-01-20	21:51:21 - 22:29:51	54.7 - 64.1	48.506	91	48.512	5.2

(continued on next page)

Table A.2 (continued).

2024-01-22	16:50:23 - 18:29:49	26.6 - 13.0	48.648	91	48.664	1.2
2024-01-22	19:05:45 - 20:50:31	17.1 - 40.0	48.662	32	48.660	3.6
2024-11-01	19:55:16 - 20:06:44	22.8 - 25.3	59.474	894	59.562	6
2024-11-01	20:08:31 - 20:15:10	25.7 - 27.3	59.423	542	59.444	3.3
2024-11-15	21:53:31 - 22:33:26	51.0 - 60.9	59.852	267	59.837	4.9
2024-11-16	19:39:52 - 20:20:35	19.9 - 28.6	59.857	150	59.853	4.5
2025-01-03	21:33:11 - 21:49:07	43.7 - 47.6	62.837	677	62.865	4.7
2025-01-09	17:19:05 - 17:48:09	27.6 - 21.5	63.496	323	63.492	6.0
2025-01-11	18:45:42 - 19:06:06	14.0 - 14.4	63.716	501	63.735	4.4
2025-01-14	17:01:53 - 17:17:44	31.5 - 27.9	64.090	488	64.078	5.0
2025-01-27	17:15:35 - 17:33:02	28.4 - 24.5	65.809	640	65.810	1.6
2025-02-10	21:59:51 - 22:18:25	47.4 - 51.9	67.968	640	67.946	1.8
2025-03-05	18:18:36 - 18:36:36	13.3 - 10.3	71.730	888	71.760	1.1
2025-06-30	21:14:17 - 21:49:08	33.2 - 41.6	97.287	776	72.968	1.6
2025-06-30	21:50:26 - 22:25:19	41.9 - 50.4	97.268	523	72.958	1.6
2025-07-11	22:53:24 - 23:37:54	57.7 - 68.7	98.515	563	73.904	1.8
1997-070A						
Observation period		Φ range	P	σ_p	P^1	Amp
Date	Time	($^{\circ}$)	(s)	(ms)	(s)	(mag)
2023-03-24	18:16:59 - 18:29:21	24.1 - 21.1	12.416	20	12.414	1.1
2023-06-29	00:33:21 - 00:47:42	76.4 - 84.6	12.263	32	12.260	4.6
2023-07-29	23:32:36 - 23:47:40	66.8 - 70.6	12.174	15	12.174	1.3
2023-09-06	22:43:31 - 22:52:54	62.6 - 65.0	12.133	49	12.130	0.8
2023-09-11	21:48:01 - 22:28:17	49.7 - 59.9	12.134	11	12.134	0.9
2023-10-06	21:06:01 - 21:55:50	44.4 - 56.9	12.132	12	12.131	0.7
2023-10-08	20:43:36 - 21:21:15	39.3 - 48.6	12.130	12	12.130	0.7
2023-10-23	19:15:25 - 21:02:34	21.5 - 46.8	12.124	10	12.110	0.7
2023-11-02	20:07:23 - 20:45:41	35.1 - 44.4	12.119	12	12.118	0.7
2023-12-05	21:12:42 - 21:31:34	55.1 - 59.7	12.105	25	12.103	1.2
2023-12-16	18:58:32 - 19:15:54	24.6 - 28.3	12.095	24	12.096	0.9
2025-01-14	17:19:08 - 17:35:06	38.5 - 34.6	11.668	19	11.668	1.6
2025-01-15	17:08:43 - 17:49:07	41 - 31.4	11.668	9	11.668	1.2
2025-01-25	20:25:19 - 20:40:23	16.2 - 18.9	11.654	17	11.657	9.1
2025-01-30	21:33:55 - 21:51:56	29.9 - 34.1	11.656	8	11.656	4.5
2025-02-04	18:50:17 - 19:09:16	18.4 - 14.9	11.649	13	11.649	5.7
2025-02-11	17:26:18 - 17:43:45	37.9 - 33.6	11.632	13	11.632	4.1
2025-02-11	18:18:04 - 18:37:01	25.5 - 21.2	11.632	12	11.632	6.3
2025-02-19	17:41:56 - 18:01:02	34.2 - 29.5	11.614	9	11.614	3.1
2025-03-05	18:37:13 - 18:54:50	20.7 - 16.6	11.585	12	11.585	1.6
2023-07-06	20:25:35 - 21:01:45	12.5 - 21.2	11.408	6	12.408	1.4

Table B.3
Best-fit parameters for the 1-panel and 4-panel models. Pole orientation is given in equatorial coordinates, bus dimensions in axis ratios (b/a , c/a), and glint-panel properties by co-latitude θ_g , azimuth ϕ_g , specular strength G , and width σ .

Parameter	1 panel model	4 panel model
pole α , $\delta[^\circ]$	331,75	344, 74
bus b/a , c/a	0.3, 2.4	0.5, 2.6
Panel 1		
$(\theta_g, \phi_g)[^\circ]$	7, 351	158, 28
G , $\sigma[^\circ]$	1, 38	10, 40
Panel 2		
$(\theta_g, \phi_g)[^\circ]$		17, 223
G , $\sigma[^\circ]$		2, 14
Panel 3		
$(\theta_g, \phi_g)[^\circ]$		24, 170
G , $\sigma[^\circ]$		1, 21
Panel 4		
$(\theta_g, \phi_g)[^\circ]$		21, 0
G , $\sigma[^\circ]$		1, 47
Total rms	0.359	0.289

References

[1] J. Zhang, Y. Cai, C. Xue, Z. Xue, H. Cai, LEO mega constellations: Review of development, impact, surveillance, and governance, *Space: Sci. Technol.* (2022).

[2] C. Pardini, L. Anselmo, Environmental sustainability of large satellite constellations in low earth orbit, *Acta Astronaut.* 170 (2020) 27–36, <http://dx.doi.org/10.1016/j.actaastro.2020.01.016>, URL <https://www.sciencedirect.com/science/article/pii/S0094576520300187>.

[3] T. Kelso, et al., Analysis of the Iridium 33-Cosmos 2251 collision, *Adv. Astronaut. Sci.* 135 (2) (2009) 1099–1112.

[4] P. Anz-Meador, J. Opiela, J.-C. Liou, History of on-orbit satellite fragmentations, Technical Report, Orbital Debris Program Office, 2023.

[5] C. Pardini, L. Anselmo, The short-term effects of the Cosmos 1408 fragmentation on neighboring inhabited space stations and large constellations, *Acta Astronaut.* 210 (2023) 465–473.

[6] C. Liang, P. Fanto, A. Signoracci, On the increased risk of kessler syndrome by anti-satellite tests, in: Proceedings of the Advanced Maui Optical and Space Surveillance (AMOS) Technologies Conference, 2023, p. 132.

[7] E. Grossman, I. Gouzman, R. Verker, Debris/micrometeoroid impacts and synergistic effects on spacecraft materials, *MRS Bull.* 35 (1) (2010) 41–47.

[8] T. Flohrer, R. Choc, B. Bastida, Classification of geosynchronous objects, 2012, GEN-DB-LOG-00086-OPS-GR.

[9] J. Šilha, J.-N. Pittet, M. Hamara, T. Schildknecht, Apparent rotation properties of space debris extracted from photometric measurements, *Adv. Space Res.* 61 (3) (2018) 844–861, <http://dx.doi.org/10.1016/j.asr.2017.10.048>.

[10] European Space Agency, ESA’s Annual Space Environment Report, Technical Report, ESA Space Debris Office, 2025, URL https://www.sdo.esoc.esa.int/environment_report/Space_Environment_Report_latest.pdf, ESA Space Debris Office, ESOC, Darmstadt, Germany.

[11] I. Gkolias, C. Colombo, Towards a sustainable exploitation of the geosynchronous orbital region, *Celest. Mech. Dyn. Astron.* 131 (4) (2019) 19.

[12] R. Longstaff, M. Hemsell, S. Alexandra, Study into the sustainable disposal of end of life GEO satellites, *J. Br. Interplanet. Soc.* 69 (2016) 429–438.

[13] C. Benson, D. Scheeres, N. Moskovitz, Light curves of retired geosynchronous satellites, in: 7th European Conference on Space Debris, Vol. 7, 2017, p. 04.

[14] C.-C.G. Chao, Accuracy of estimating solar radiation pressure for GEO debris with tumbling effect, in: H. Lacoste (Ed.), Fifth European Conference on Space Debris, in: ESA Special Publication, Vol. 672, 2009, p. 122.

[15] P. Papushev, Y. Karavaev, M. Mishina, Investigations of the evolution of optical characteristics and dynamics of proper rotation of uncontrolled geostationary artificial satellites, *Adv. Space Res.* 43 (9) (2009) 1416–1422.

[16] C.R. Binz, M.A. Davis, B.E. Kelm, C.I. Moore, Optical survey of the tumble rates of retired GEO satellites, in: Advanced Maui Optical and Space Surveillance Technologies Conference, Vol. 1, Curran Associates Inc. New York, 2014, p. 61.

- [17] J. Africano, P. Kervin, D. Hall, P. Sydney, J. Ross, T. Payne, S. Gregory, K. Jorgensen, K. Jarvis, T. Parr-Thumm, et al., Understanding photometric phase angle corrections, in: 4th European Conference on Space Debris, Vol. 587, 2005, p. 141.
- [18] G.W. Ojakangas, H. Cowardin, N. Hill, Toward realistic dynamics of rotating orbital debris, and implications for lightcurve interpretation, in: Advanced Maui Optical and Space Surveillance Technologies Conference (AMOS 2011), 2011, JSC-CN-24227.
- [19] S. Fan, C. Frueh, A direct light curve inversion scheme in the presence of measurement noise, *J. Astronaut. Sci.* 67 (2) (2020) 740–761, <http://dx.doi.org/10.1007/s40295-019-00190-3>.
- [20] M.A. Earl, G.A. Wade, Observations of the spinperiod variations of inactive box-wing geosynchronous satellites, *J. Spacecr. Rockets* 52 (3) (2015) 968–977.
- [21] M. Earl, G. Wade, The long-term broadband photometric surveys of inactive AS-7000 and HS-601 box-wing geosynchronous satellites, in: Proceedings of the Canadian Aeronautical and Space Institute (CASI) ASTRO2016 Conference, Royal Military College of Canada, Ottawa, ON, Canada, 2016, Presented on May 18, 2016.
- [22] M.A. Earl, P.W. Somers, K. Kabin, D. Bédard, G.A. Wade, Estimating the spin axis orientation of the Echostar-2 box-wing geosynchronous satellite, *Adv. Space Res.* 61 (8) (2018) 2135–2146.
- [23] J. Šilhá, J.-N. Pittet, M. Hamara, T. Schildknecht, Apparent rotation properties of space debris extracted from photometric measurements, *Adv. Space Res.* 61 (3) (2018) 844–861.
- [24] M. Birlan, A. Sonka, O. Badescu, A. Nedelcu, P. Paraschiv, V. Turcu, S. Anghel, D. Besliu-Ionescu, I. Boaca, M. Trelia, et al., Berthelot, the new astronomical station in Romania, *Rom. Astron. J.* 31 (1) (2021) 41–56.
- [25] C.A. Danescu, O. Blagoi, L. Dumitru, The new remote solar observatory at general berthelot, *Rom. Astron. J.* 34 (3) (2024) 123–136.
- [26] P. Qiu, Y. Zhao, J. Zheng, J.-F. Wang, X.-J. Jiang, Research on performances of back-illuminated scientific CMOS for astronomical observations, *Res. Astron. Astrophys.* 21 (10) (2021) 268, <http://dx.doi.org/10.1088/1674-4527/21/10/268>.
- [27] D.J. McKay, P. Ballester, K. Banse, C. Izzo, Y. Jung, M. Kiesgen, N. Kornweibel, L.K. Lundin, A. Modigliani, R.M. Palsa, C. Sabet, The common pipeline library: standardizing pipeline processing, in: P.J. Quinn, A. Bridger (Eds.), Optimizing Scientific Return for Astronomy Through Information Technologies, in: Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series, Vol. 5493, 2004, pp. 444–452, <http://dx.doi.org/10.1117/12.551214>.
- [28] C.P. Hernández, K. Cáceres Mosqueira, C. Jobic, F.V. Sallés, A.M. Antón, D. Moreno-Espinosa, EUSST sensor calibration procedure, in: Proceedings of the Advanced Maui Optical and Space Surveillance (AMOS) Technologies Conference, 2023, p. 114.
- [29] Inter-Agency Space Debris Coordination Committee, IADC space debris mitigation guidelines, 2020, <https://www.iadc-home.org>, IADC-02-01, Revision 2.
- [30] J. Bangert, W. Puatua, G. Kaplan, J. Bartlett, W. Harris, A. Fredericks, A. Monet, User's guide to NOVAS version C3.1, Technical Report, United States Naval Observatory, Washington, DC, 2011, USNO Technical Report.
- [31] D.A. Vallado, P. Crawford, R. Hujsak, T. Kelso, Revisiting spacetrack report #3: Rev 2, in: AIAA Astrodynamics Specialist Conference Paper, 2006, AIAA-2006-6753.
- [32] B. Gray, sat_code: C implementation of SGP4/SDP4 satellite motion models, 2019, Project Pluto, GitHub repository, https://github.com/Bill-Gray/sat_code.
- [33] K. Barbary, SEP: Source extraction and photometry, 2018, Astrophysics Source Code Library, record ascl:1811.004.
- [34] E. Bertin, S. Arnouts, SExtractor: Software for source extraction, *Astron. Astrophys. Suppl.* 117 (1996) 393–404, <http://dx.doi.org/10.1051/aas:1996164>.
- [35] S.B. Howell, Two-dimensional aperture photometry: signal-to-noise ratio of point-source observations and optimal data-extraction techniques, *Publ. Astron. Soc. Pac.* 101 (640) (1989) 616, <http://dx.doi.org/10.1086/132477>.
- [36] G.H. Kaplan, et al., Explanatory supplement to the astronomical almanac, *Astron. J.* 97 (1989) 1197–1210.
- [37] G.H. Kaplan, et al., Chapter 3, in: Explanatory Supplement To the Astronomical Almanac, University Science Books, 1992, pp. 95–198.
- [38] F.G. Valdes, L.E. Campusano, J.D. Velasquez, P.B. Stetson, FOCAS automatic catalog matching algorithms, *Publ. ASP* 107 (1995) 1119, <http://dx.doi.org/10.1086/133667>.
- [39] Consultative Committee for Space Data Systems (CCSDS), Tracking data message (TDM), recommended standard, issue 1, Technical Report CCSDS 503.0-B-1, CCSDS, 2007.
- [40] M. Zechmeister, M. Kürster, The generalised Lomb-Scargle periodogram. A new formalism for the floating-mean and Keplerian periodograms, *Astron. Astrophys.* 496 (2) (2009) 577–584, <http://dx.doi.org/10.1051/0004-6361:200811296>.
- [41] J.T. VanderPlas, Understanding the lomb-scargle periodogram, *Astrophys. J. Suppl. Ser.* 236 (1) (2018) 16.
- [42] R.F. Stellingwerf, Period determination using phase dispersion minimization, *Astrophys. J.* 224 (1978) 953–960, <http://dx.doi.org/10.1086/156444>.
- [43] A. Rachman, A. Vananti, T. Schildknecht, Spin period evolution of decommissioned GLONASS satellites, *Aerospace* 12 (4) (2025) <http://dx.doi.org/10.3390/aerospace12040283>, URL <https://www.mdpi.com/2226-4310/12/4/283>.
- [44] N. Cimmino, A. Vananti, T. Schildknecht, Comparing the spin period evolution of inactive box-wing satellites and rocket bodies, in: Proceedings of the 11th European Conference for AeroSpace Sciences, EUCAST, 2025, ADD_DOI_HERE, Conference dates: 30 June–4 July 2025. Corresponding author: Nicola Cimmino.
- [45] E. Bowell, B. Hapke, D. Domingue, K. Lumme, J. Peltoniemi, A.W. Harris, Application of photometric models to asteroids, in: R.P. Binzel, T. Gehrels, M.S. Matthews (Eds.), Asteroids II, University of Arizona Press, Tucson, AZ, USA, 1989, pp. 524–556.
- [46] P. Magnusson, M.A. Barucci, J.D. Drummond, K. Lumme, S.J. Ostro, J. Surdej, R.C. Taylor, V. Zappalà, Determination of pole orientations and shapes of asteroids, in: R.P. Binzel, T. Gehrels, M.S. Matthews (Eds.), Asteroids II, University of Arizona Press, Tucson, AZ, 1989, pp. 66–97.
- [47] M. Kaasalainen, J. Torppa, Optimization methods for asteroid lightcurve inversion. I. Shape determination, *Icarus* 153 (1) (2001) 24–36, <http://dx.doi.org/10.1006/icar.2001.6673>.
- [48] M. Kaasalainen, J. Torppa, K. Muinonen, Optimization methods for asteroid lightcurve inversion. II. The complete inverse problem, *Icarus* 153 (1) (2001) 37–51, <http://dx.doi.org/10.1006/icar.2001.6674>.
- [49] D.O. Fulcoy, K.I. Kalamaroff, F.K. Chun, Determining basic satellite shape from photometric light curves, *J. Spacecr. Rockets* 49 (1) (2012) 76–82, <http://dx.doi.org/10.2514/1.A32002>.
- [50] F. Piergentili, F. Santoni, P. Seitzer, Attitude determination of orbiting objects from lightcurve measurements, *IEEE Trans. Aerosp. Electron. Syst.* 53 (1) (2017) 81–90.
- [51] D. Hall, B. Calef, K. Knox, M. Bolden, P. Kervin, Separating attitude and shape effects for non-resolved objects, in: The 2007 AMOS Technical Conference Proceedings, Maui Economic Development Board, Inc. Kihei, Maui, HI, 2007, pp. 464–475.
- [52] A. Burton, C. Frueh, Two methods for light curve inversion for space object attitude determination, in: 8th European Conference on Space Debris, 2021.
- [53] L. Robinson, C. Frueh, Light curve inversion for reliable shape reconstruction of human-made space objects, in: Proceedings of the 32nd AIAA/AAS Astrodynamics Specialist Conference, AIAA Reston, VA, 2022, pp. 1–19.
- [54] R. Linares, J.L. Crassidis, M.K. Jah, Particle filtering light curve based attitude estimation for non-resolved space objects, in: Aas/Aiaa Space Flight Mechanics Meeting, 2014, pp. 14–210.
- [55] D.V. Cabrera, J. Utmann, R. Förstner, The adaptive Gaussian mixtures unscented Kalman filter for attitude determination using light curves, *Adv. Space Res.* 71 (6) (2023) 2609–2628.
- [56] R. Clark, S. Dave, J. Wawrow, R. Lee, Performance of parameterization algorithms for resident space object (rso) attitude estimates, in: Proceedings of the Advanced Maui Optical and Space Surveillance Technologies, Maui, HI, USA, 2020, pp. 15–18.
- [57] R. Clark, Y. Fu, S. Dave, R.S. Lee, Resident space object (RSO) attitude and optical property estimation from space-based light curves, *Adv. Space Res.* 70 (11) (2022) 3271–3280.
- [58] A. Burton, L. Robinson, C. Frueh, Light curve attitude estimation using particle swarm optimizers, *Adv. Space Res.* 74 (11) (2024) 5619–5638.
- [59] K. Lumme, E. Bowell, Radiative transfer in the surfaces of atmosphereless bodies. I – Theory, *Astron. J.* 86 (1981) 1694–1704, <http://dx.doi.org/10.1086/113054>.
- [60] B. Hapke, Theory of Reflectance and Emittance Spectroscopy, Cambridge University Press, Cambridge, 1993.
- [61] M. Kaasalainen, Interpretation of lightcurves of precessing asteroids, *Astron. Astrophys.* 376 (1) (2001) 302–309.
- [62] R. Storn, K. Price, Differential evolution – a simple and efficient heuristic for global optimization over continuous spaces, *J. Global Optim.* 11 (4) (1997) 341–359, <http://dx.doi.org/10.1023/A:1008202821328>.
- [63] J.J. Moré, The Levenberg-Marquardt algorithm: Implementation and theory, in: G.A. Watson (Ed.), Numerical Analysis, Springer Berlin Heidelberg, Berlin, Heidelberg, 1978, pp. 105–116.
- [64] P. Virtanen, R. Gommers, T.E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S.J. van der Walt, M. Brett, J. Wilson, K.J. Millman, N. Mayorov, A.R.J. Nelson, E. Jones, R. Kern, E. Larson, C. Carey, Í. Polat, Y. Feng, E.W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E.A. Quintero, C.R. Harris, A.M. Archibald, A.H. Ribeiro, F. Pedregosa, P. van Mulbregt, SciPy 1.0 Contributors, SciPy 1.0: fundamental algorithms for scientific computing in Python, *Nature Methods* 17 (2020) 261–272, <http://dx.doi.org/10.1038/s41592-019-0686-2>.